



A review of recent advances and strategies in transfer learning

Masoume Gholizade¹ · Hadi Soltanizadeh¹ · Mohammad Rahmanimanesh¹ ·
Shib Sankar Sana²

Received: 18 April 2024 / Revised: 7 December 2024 / Accepted: 30 December 2024 / Published online: 21 February 2025

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Abstract While machine learning has achieved remarkable success across various applications, this success often relies on the assumption that training and testing data originate from the same domain with a shared distribution and feature space. This assumption is frequently violated in real-world scenarios where training data may derive from different distributions, often due to limited access to labeled samples or data collection challenges. This situation necessitates the development of robust methodologies encapsulated in the framework of transfer learning, which aims to enhance learner performance in different domains by leveraging knowledge from a source domain to improve outcomes in a target domain. This review provides a comprehensive assessment of the evolving landscape of transfer learning, highlighting the need for a detailed literature survey on current approaches and their implications. By categorizing theoretical transfer learning algorithms based on the number of source domains and the handling of incomplete multi-source data, we offer a distinctive perspective that elucidates the practical utility of these methodologies. Our exploration spans various pivotal categories, including single-source/single-target and multi-source/single-target

frameworks, while systematically addressing the complexities of incomplete transfer learning and the integration of fuzzy systems to manage uncertainty. Additionally, we introduce commonly used public datasets in transfer learning and present the outcomes of several key algorithms discussed in this review on these datasets. This empirical perspective illustrates the practical effectiveness of the methodologies and provides insights into their performance under diverse conditions. This extensive analysis summarizes key findings from diverse studies, delineates their interconnections, and highlights their applicability to real-world problems, thereby serving as a valuable resource for researchers navigating the intricate dynamics of transfer learning.

Keywords Transfer learning · Machine learning · Multi-source learning · Single-source learning · Inductive learning · Transductive learning · Fuzzy systems · Incomplete data · Literature review

1 Introduction

Machine learning (ML) is a subset of artificial intelligence that enables systems to learn from experience without being explicitly programmed automatically. It focuses on developing algorithms that can analyze data, recognize patterns, and make decisions or predictions based on that data. Unlike rule-based systems, which rely on predefined instructions, machine learning algorithms adapt as they are exposed to new data, improving their performance over time. The growing availability of large datasets, combined with advances in computational power, has accelerated the adoption of machine learning across various industries, making it an indispensable tool for solving complex problems in finance, healthcare, and autonomous systems (Kadhim 2019).

✉ Shib Sankar Sana
shib_sankar@yahoo.com

Masoume Gholizade
masoume_gholizade@semnan.ac.ir

Hadi Soltanizadeh
h_soltanizadeh@semnan.ac.ir

Mohammad Rahmanimanesh
rahmanimanesh@semnan.ac.ir

¹ Faculty of Electrical and Computer Engineering, Semnan University, Semnan, Iran

² Department of Mathematics, Kishore Bharati Bhagini Nivedita College, Behala, Kolkata 700060, India

Significant advancements have been made in the field of machine learning, enabling its application in a variety of areas, such as object detection (Chen et al. 2018a), speech recognition (Asaei et al. 2017), face and position recognition (Khaleghi et al. 2022), image processing (Zhong et al. 2019), and natural language processing (Zhao et al. 2014). Machine learning techniques have effectively addressed challenges that traditional statistical methods could not overcome (Niu et al. 2021). These algorithms have shown impressive results across many domains, proving their efficacy in predicting future events based on learned patterns from training data (Huang et al. 2020).

Most traditional machine learning models, such as Principal Component Analysis (PCA) (Deutsch 2002), Naïve Bayes (Rish 2001), Support Vector Machines (SVM) (Cortes and Corinna (AT&T Bell Labs., Hightstown, NJ 07933 and U. Vladimir, Vapnik (AT&T Bell Labs., Hightstown, NJ 07933, “Support-Vector Networks”, *Mach. Learn.*, 1995, 297(20), pp. 273–297. 1995), Hidden Markov Models (HMM) (Balakrishna and Ganapathiraju xxxx), K-Nearest Neighbors (KNN), Logistic Regression (LR), and Linear Discriminant Analysis (LDA) (Shawe-Taylor and Sun 2011), operate under the assumption that the training and test data are drawn from the same distribution and share identical feature spaces. However, the performance of these models is contingent on having sufficient labeled data for training (Sun et al. 2015). Specifically, traditional machine learning requires large quantities of labeled data collected in the same distribution as the test data to build a robust model (Wang et al. 2019a). For example, in many real-world car detection systems, training and test images must be captured under similar conditions—using the same camera, positioned in the same location, under comparable lighting and clarity conditions—to ensure the model performs accurately (Hasanvand 2023).

Conventional machine learning approaches face three primary challenges: insufficient labeled data, mismatched feature spaces, and incompatible training and test data distributions. These challenges often lead to suboptimal performance when applying these models to new datasets. With insufficient labeled data, practitioners may use data from other sources, leading to conflicts in feature spaces and distributions, further degrading model performance. Real-world examples where the assumption of identical feature spaces and distributions does not hold include scenarios like cross-device object detection or multilingual speech recognition, where the training and testing data come from different devices or languages, leading to significant performance drops (Pan and Yang Oct. 2010).

For example, in medical imaging, a model trained on MRI scans from one hospital may not generalize well to MRI scans from another hospital due to differences in imaging equipment, patient demographics, and scan protocols. These

discrepancies in data distributions lead to performance drops, illustrating that the assumption of the same feature space and distribution is often unrealistic in real-world applications (Khaleghi et al. 2022). Similarly, cross-language speech recognition or cross-device object detection often reduces accuracy when training and test data come from different sources or sensors (Chen et al. 2023).

Another practical example can be observed in remote sensing (Chen et al. 2022a). Consider the scenario illustrated in Fig. 1, where an airplane captures images of the same geographical area from two different positions, A and B. These positions differ in altitude, orientation, and lighting conditions. These variations create domain shifts, meaning the data distributions between images captured at positions A and B differ significantly. When a model is trained solely on images from position A, it may struggle to maintain high performance when applied directly to images from position B. This drop in accuracy is due to the model's inability to generalize across the domain shift caused by changes in perspective, illumination, and other environmental factors. However, transfer learning techniques can be applied to address this issue. By leveraging knowledge acquired from training on position A, the model can adapt to position B. This process involves transferring the learned representations and fine-tuning the model using domain adaptation strategies. These methods align the feature spaces or reduce distribution discrepancies between the two domains, ultimately enhancing the model's ability to classify or analyze images from position B with improved accuracy and robustness. This example highlights the practical importance of domain adaptation in real-world applications, where domain shifts often occur due to varying conditions during data collection.

To overcome these challenges, transfer learning has emerged as a powerful technique. Transfer learning leverages knowledge gained from one task (the source domain) and applies it to a different but related task (the target domain), even when the two domains differ in feature space or data distribution (Pan et al. 2011). Transfer learning reduces the need for extensive labeled data in the target domain by utilizing pre-learned features and models from the source domain. This capability is particularly valuable when obtaining large amounts of labeled data is costly, time-consuming, or impractical.

The accompanying image (Fig. 2) visually represents the transfer learning process between two domains. In the source domain (upper part of Fig. 2), a model is trained on a set of objects such as a bicycle, a monitor, and a calculator. This represents the initial task, where there is typically sufficient labeled data for the model to learn patterns and features. Once the model has been trained on this source domain, the learned knowledge (e.g., features or model weights) is transferred to a new target domain (lower part of Fig. 2), where similar but slightly different objects are presented,

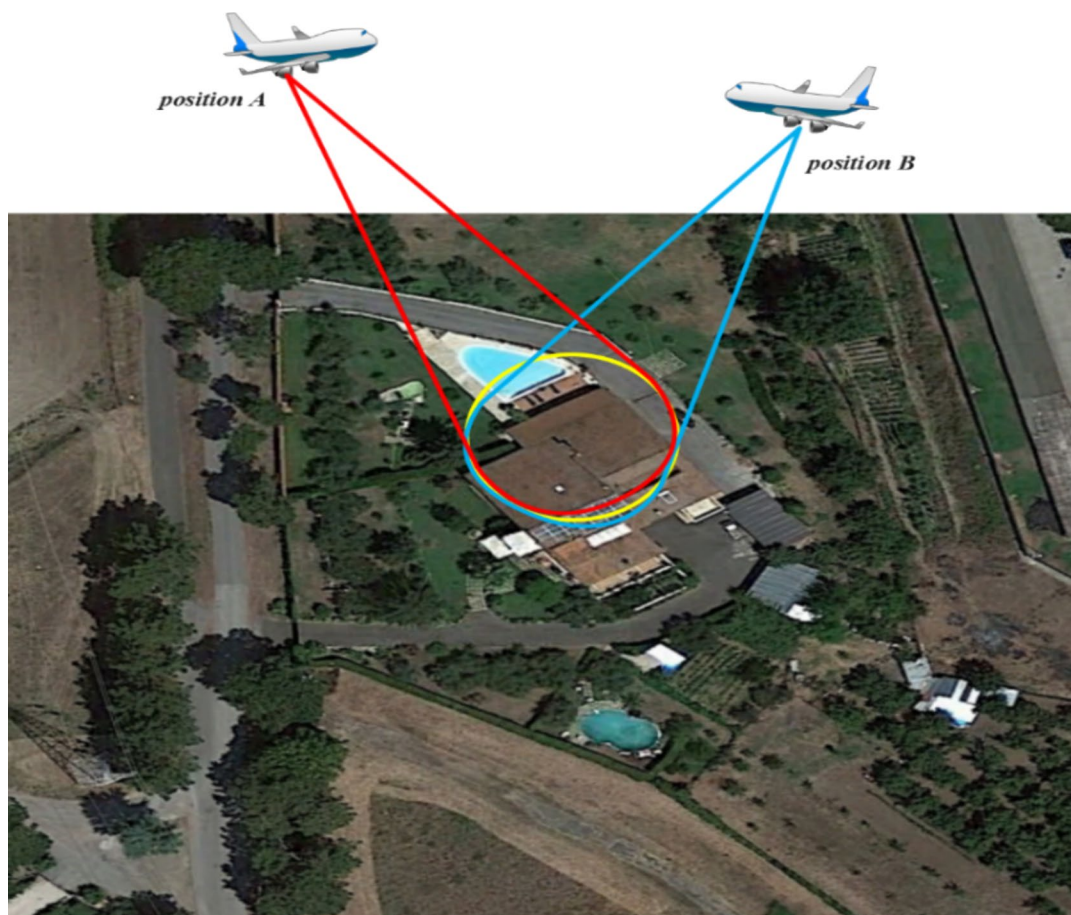


Fig. 1 Illustration of domain shifts in remote sensing image classification. as an airplane captures images of the same geographic area from different positions (a and b), variations in altitude, orientation, and sensor angles lead to changes in the spatial and spectral properties of the images

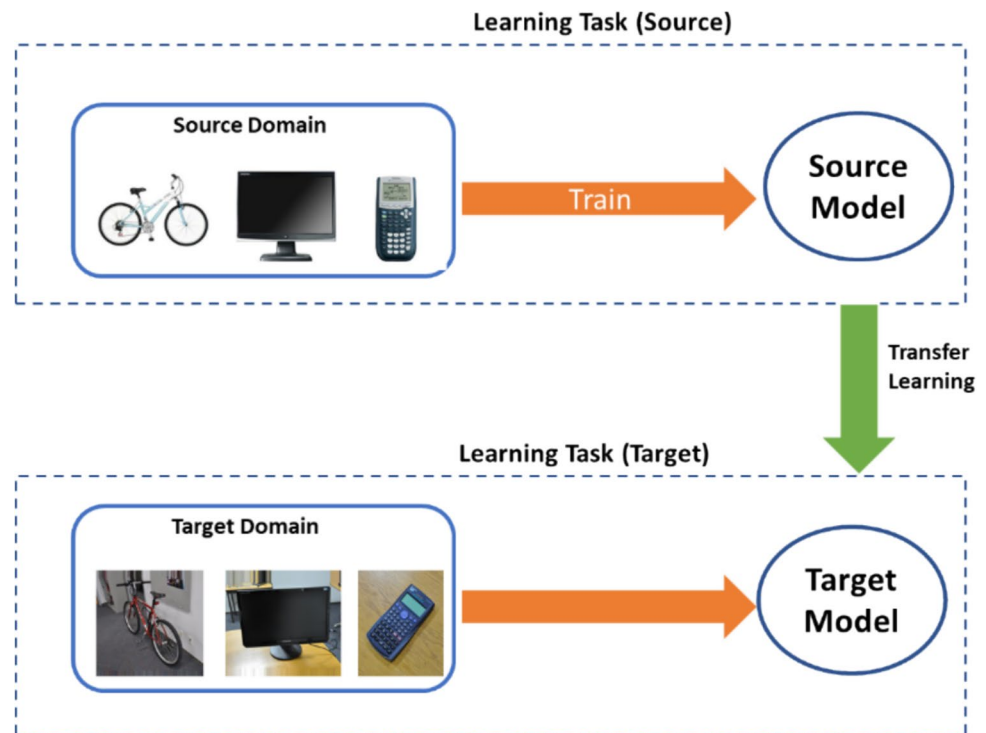
such as another bicycle, monitor, and calculator. However, the target domain may have less labeled data or a different data distribution compared to the source domain. The green arrow between the two domains indicates the transfer of knowledge from the source domain to the target domain. By transferring this knowledge, the model can perform well in the target domain without the need for extensive retraining or large amounts of labeled data. The second orange arrow represents how this transferred knowledge is applied or fine-tuned for the new task in the target domain. This process reduces the need for collecting and labelling a vast amount of data in the target domain, making transfer learning a highly valuable approach in scenarios where data is scarce or difficult to obtain. As a result, transfer learning enables the development of efficient and adaptable models to different but related tasks, which is crucial in fields like computer vision, natural language processing, and healthcare.

In a typical transfer learning framework, a model is initially pre-trained on a large-scale dataset from a source domain. This pre-training phase allows the model to learn generalizable features and representations. Subsequently,

the pre-trained model is fine-tuned or adapted to achieve optimal performance on a smaller, domain-specific dataset from the target domain. For example, a model trained on an extensive dataset of general medical images can be fine-tuned to detect rare diseases using a smaller, specialized dataset. The model effectively transfers its learned representations by leveraging the knowledge acquired during the pre-training phase, enabling improved performance in the target domain. This approach is particularly valuable when labeled data in the target domain is scarce, as it mitigates the challenges associated with data insufficiency while maintaining high classification or detection accuracy (Pan et al. 2011).

Multiple studies have supported claims about transfer learning's ability to function with less training data. In particular, transfer learning has been shown to reduce the need for labeled data in areas such as image recognition, natural language processing, and medical diagnosis, where labeled data is often scarce. This approach improves model performance and reduces the time and cost associated with data collection and annotation.

Fig. 2 Transfer Learning Scenario



The benefits of transfer learning extend beyond its ability to function with less training data. It enables models to adapt more efficiently to new environments, tasks, or domains. It is highly applicable in real-world settings where data collection can vary significantly between regions, devices, or populations. For example, models trained on large, diverse datasets in healthcare can be fine-tuned for specialized tasks like rare disease detection, where data is often insufficient (Weiss et al. 2016).

In recent years, the attention on transfer learning has increased significantly due to its effectiveness in addressing various challenges across different domains. The growing body of literature on transfer learning reflects its relevance and adaptability, with a noticeable rise in publications on this topic. For instance, as illustrated in the following chart, the number of research articles published on transfer learning has surged over the past few years, indicating a burgeoning interest in this study area (Pan and Yang Oct. 2010).

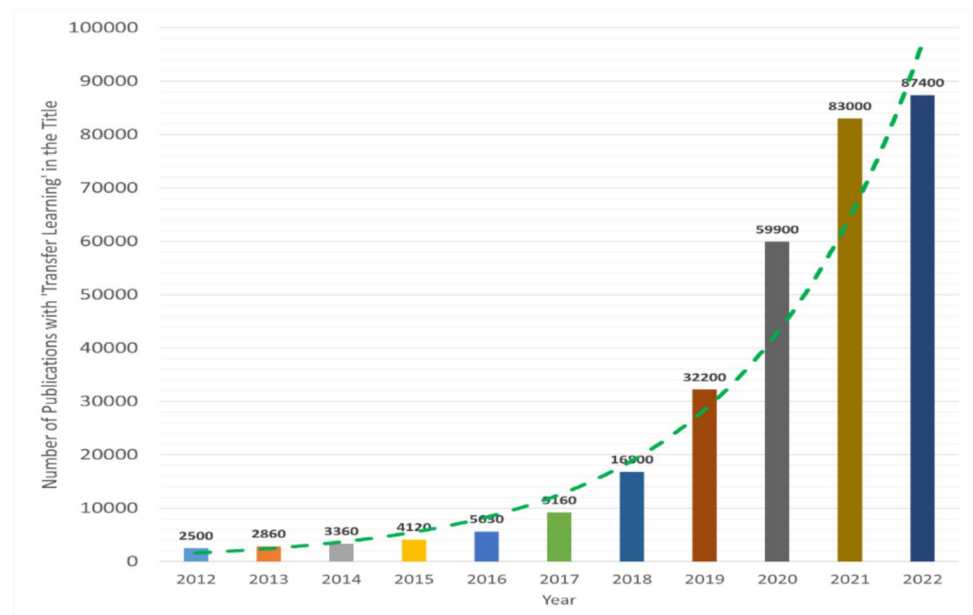
Figure 3 illustrates the growth of academic publications that feature the term "transfer learning" in their titles over the years. The horizontal axis represents the years, ranging from the initial studies in this field to the present day, while the vertical axis quantifies the number of publications each year. Notably, a significant upward trend indicates an increasing interest and investment in transfer learning research. This surge in publications reflects the growing recognition of transfer learning as a critical approach for addressing complex challenges in various domains, including computer vision, natural language processing, and

medical diagnostics. The data for this analysis was gathered from Google Scholar, underscoring how researchers increasingly leverage transfer learning techniques to enhance model performance, particularly in scenarios where labeled data is scarce or mismatches in feature spaces and distributions. This trend not only underscores the relevance of transfer learning in contemporary research but also points to its potential for future advancements in artificial intelligence.

Despite its promising applications, transfer learning also presents challenges that need to be addressed. For instance, the lack of certainty in the transfer process highlights the effectiveness of fuzzy systems in solving transfer learning problems, particularly in scenarios where data is imprecise or uncertain (Zuo et al. 2019a; Liu et al. 2021a). Additionally, in some multi-source transfer learning scenarios, accessing samples from each source may not be feasible, leading to situations where only the model from the source domain is available for adaptation (Gong et al. 2020; Ding et al. 2018; Lu et al. 2020a). These challenges necessitate further exploration of methodologies that can enhance the efficacy of transfer learning in diverse applications.

The primary purpose of this paper is to provide a comprehensive survey of transfer learning approaches by categorizing existing research and establishing connections across different methodologies. We aim to offer a structured understanding of transfer learning by focusing on key dimensions such as the number of information sources and the application of fuzzy systems to handle uncertainty [27]. Through this survey, we not only summarize and analyze significant

Fig. 3 Number of Publications with the Term "Transfer Learning" in Their Titles Over the Years



advancements in the field but also shed light on challenges that have yet to be fully addressed, such as the limitations of multi-source learning and the need for more robust transfer learning methods in scenarios with incomplete data.

To accomplish this, we first collected relevant publications from the last decade using keywords such as "transfer learning," "homogeneous transfer learning," and "multi-source transfer learning" in Google Scholar. After filtering through these studies, we organized them according to key themes aligned with the goals of this survey. For instance, we categorize transfer learning based on the availability of labels in the source and target domains, dividing it into transductive, inductive, and unsupervised learning, as illustrated in Fig. 3. This categorization reflects the varying degrees of

supervision and the different challenges researchers face, depending on the availability and nature of labels in the data.

In Fig. 4, we further delineate transfer learning by examining its classification within two key dimensions: the label and feature spaces. The label space includes three categories: inductive, transductive, and unsupervised learning, which Fig. 5 correspond to the degree of supervision available in the source and target domains. Meanwhile, the feature space distinguishes between homogeneous and heterogeneous transfer learning, highlighting the importance of similarity in feature representation for effective knowledge transfer. Homogeneous transfer learning occurs when the source and target domains share the same feature space, allowing for straightforward

Fig. 4 Transfer Learning Categories

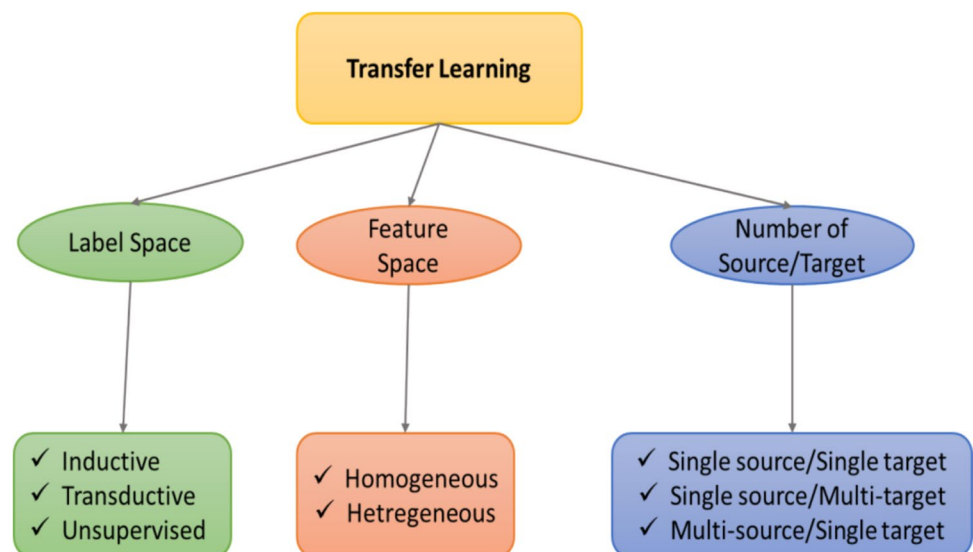
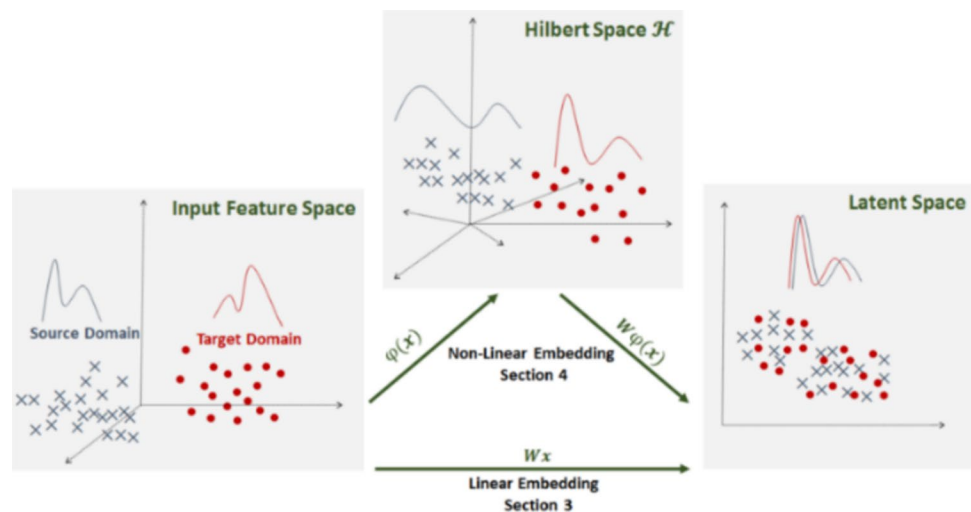


Fig. 5 The scheme of DME (Baktashmotlagh et al. 2016)



knowledge transfer. This scenario generally leads to more effective learning, as the model can directly apply knowledge gained from the source to the target without significant transformation. Heterogeneous transfer learning, on the other hand, involves different feature spaces for the source and target domains. This necessitates advanced feature transformation or alignment methods, which complicates the transfer process and opens avenues for richer knowledge integration (Gholizade et al. 2021).

Furthermore, we categorize approaches based on the number of domains involved: single-source/single-target, multi-source/single-target, and single-source/multi-target scenarios. This classification underscores the complexity of transfer learning and the various strategies available to researchers, as it influences the design and applicability of transfer learning models in diverse situations.

We also explore prior categorizations, such as those proposed in Pan and Yang Oct. (2010), which break down transfer learning into feature-based (Pan et al. 2010; Fernando et al. 2013), instance-based (Yao and Doretto 2010; Chattopadhyay et al. 2012; Chen et al. 2019), relational-based (Zuo et al. 2019a; Liu et al. 2021a), and parameter-based approaches. These categories highlight the diversity of problem-solving strategies in the field, ranging from transforming feature representations to transferring knowledge at the model or parameter level (Zhao et al. 2011; Long et al. 2017a; Deng et al. 2014a; Li et al. 2012).

In comparison to previous surveys, our paper introduces several significant contributions:

This paper reviews over 50 studies conducted since 2015, providing a detailed examination of key findings and contributions within transfer learning. We complement theoretical discussions with practical experiments involving

over 15 transfer learning models to illustrate performance variations across different approaches.

Each category of transfer learning is discussed in-depth, with an emphasis on multi-source domains. We scrutinize the interconnections between results from different subfields, offering a holistic perspective on the evolution of transfer learning methodologies.

Our survey ventures into underexplored topics, such as incomplete transfer learning and fuzzy systems, areas largely absent in recent reviews. This paper provides a forward-looking perspective on transfer learning's potential by tackling emerging strategies.

The remaining sections of this paper are organized as follows: Sect. 2 provides a comprehensive review of various survey articles on transfer learning, highlighting the significant advancements and improvements made in our study compared to previous works. Section 3 presents fundamental definitions and core concepts associated with transfer learning, offering a foundational understanding of the subject matter that will underpin the discussions in the subsequent sections. Section 4 elaborates on the different transfer learning methods and categories, detailing the various approaches utilized in the field and their respective applications in addressing challenges related to data distribution and feature space differences. Section 5 focuses on publicly available datasets and performance evaluation metrics commonly employed in transfer learning studies, showcasing the empirical data used for validating the proposed methods and the metrics that assess their effectiveness. Finally, Sect. 6 concludes the paper by summarizing key insights derived from the study and providing recommendations for future research directions,

aiming to synthesize findings and suggest pathways for advancing the field of transfer learning.

2 Related work

Transfer learning has emerged as a powerful technique for reusing knowledge across different domains, addressing critical challenges like data scarcity, distribution mismatches, and computational limitations. Over the last decade, various surveys have reviewed transfer learning's developments, each focusing on unique aspects such as domain adaptation, source-free adaptation, and knowledge distillation. However, these surveys often leave gaps in exploring emerging areas such as incomplete transfer learning, fuzzy transfer learning, and the complex dynamics of multi/single-source and multi/single-target domains. In this section, we will review major works while emphasizing the unique contributions of our paper, particularly in terms of fuzzy transfer learning and multi/single-source and multi/single-target domains.

One of the foundational surveys in the field is by Niu et al. (Niu et al. 2021), which provides a decade-long review of transfer learning, categorizing approaches into transductive, inductive, unsupervised, and negative transfer learning frameworks. While this survey provides a high-level overview and discusses ethical concerns such as data privacy and job displacement, it lacks a deeper exploration of specific subfields like multi-source transfer learning and fuzzy transfer learning. Our survey fills this gap by focusing on how fuzzy logic can enhance transfer learning models, especially in multi-source to multi-target settings where data ambiguity and uncertainty are prevalent.

Yu et al. (Li et al. 2024) offer a focused review of domain adaptation, particularly highlighting Source-Free Domain Adaptation (SFDA). Their work is highly relevant for applications where sharing sensitive data is infeasible due to privacy or security concerns. SFDA allows models to adapt from one domain to another without requiring access to source data. While their survey emphasizes source-free methods, it overlooks the potential of fuzzy transfer learning in handling uncertainty in real-world applications. Our review discusses multi-source and fuzzy transfer learning, showing how the latter can be applied to complex, uncertain environments across multiple domains.

Another major survey by Weiss et al. (Weiss et al. 2016) concentrates on homogeneous transfer learning, offering an extensive overview of transfer learning approaches for text categorization and object recognition. Their review systematically covers over 40 transfer learning methods but is largely limited to cases with full data availability. It does not address incomplete or fuzzy transfer learning scenarios

where data may be partially missing or uncertain. Our paper addresses this limitation by introducing fuzzy transfer learning strategies, showing how they can be applied when only partial or noisy data is available, as is common in IoT and healthcare applications.

Pan et al. (Pan and Yang Oct. 2010) present an overview of transfer learning's real-world challenges, such as data scarcity and computational inefficiencies in areas like image processing, speech recognition, and NLP. However, their review focuses on traditional transfer learning approaches and lacks a thorough examination of multi-source/target transfer and fuzzy transfer learning. In contrast, we delve into these areas, highlighting the advantages of multi-source transfer learning methods in leveraging information from multiple domains and how fuzzy logic can deal with uncertain or ambiguous data in these scenarios.

A recent contribution to the field by Liu et al. (Liu et al. 2021b) focuses on data-free knowledge transfer, particularly addressing knowledge distillation and source-free domain adaptation. Knowledge distillation involves transferring intra-domain knowledge from a cumbersome teacher model to a more compact student model for purposes such as model compression and efficient inference. At the same time, SFDA adapts a source model to a target domain without accessing the original training data. This survey connects the two under a unified data-free knowledge transfer framework, offering a hierarchical taxonomy and formal definitions for both areas. While this work provides a comprehensive review of data-free approaches, it doesn't address fuzzy transfer learning or multi/single-source transfer to multi/single-target domains. Our paper extends this discussion by exploring how fuzzy logic can enhance data-free methods, allowing them to handle incomplete or ambiguous data in multi-source/target scenarios.

In conclusion, while these surveys offer valuable insights into various aspects of transfer learning, they leave significant gaps in fuzzy transfer learning and the complexities of multi/single-source transfer to multi/single-target domains. Our review fills these gaps by thoroughly analysing these underexplored areas. It demonstrates how they can solve real-world problems where uncertainty, incomplete knowledge, and data privacy concerns are key issues. We also extend the discussion to data-free knowledge transfer methods, integrating them into our broader fuzzy and multi-source transfer learning framework.

3 Preliminaries

Machine learning models often rely on the assumption that the training and testing data are drawn from the same distribution and share a common feature space. However, this

Table 1 Comparison notations in transfer learning

symbol	description	symbol	description
$\mathcal{X}$	Feature space	n	Number of instances
$\mathcal{Y}$	Label space	X	Instance set
N	Feature dimensional	$P(X)$	Marginal distribution
Y	Label set corresponding to X	$p(y X)$	Conditional distribution
$\mathcal{T}$	Learning task	$f(x)$	decision function
U	Unlabeled instances	L	Labeled instances
Φ	Mapping function	$\mathcal{H}$	Reproducing kernel Hilbert space
<i>Subscripts</i>	Source	$\mathcal{D}_s$	Target domain source
<i>Subscriptt</i>	target	$\mathcal{D}_t$	Target domain data

assumption is frequently violated in real-world applications due to differences in the data distribution between domains. Transfer learning provides a solution by enabling knowledge transfer from a source domain to a target domain, improving model performance in the target domain even when limited labeled data is available.

In transfer learning, domains can vary regarding their feature space, marginal distributions, or both. The goal is to use the knowledge acquired from one or more source domains to enhance learning in a target domain, overcoming the limitations imposed by data scarcity or domain divergence. This section introduces some fundamental concepts and notations for transfer learning. It overviews key categories, including single-source and multi-source transfer learning, the integration of fuzzy systems for uncertainty management, and the challenges associated with incomplete transfer learning.

Table 1 provides a set of standard notations for transfer learning to ensure consistency throughout this review. Subsequent sections will reference these notations.

Domain: In Domain $\mathcal{D}$, there are two key aspects: $\mathcal{X}$, which defines the feature space with N dimensions, and P , which represents the marginal distribution for the instance set X , denoted as $X = \{x_1, x_2, \dots, x_n\} \in \mathcal{X}$. Consequently, Domain $\mathcal{D}$ can be represented as $\mathcal{D} = \{\mathcal{X}, P(X)\}$. As per this definition, disparities between the source and target domains, $\mathcal{D}_s \neq \mathcal{D}_t$, arise from variations in the feature space and the distribution of features (Liu et al. 2021b).

Task: Task $\mathcal{T}$ contains two parameters: parameter $\mathcal{Y}$ as label space and parameter $f(\cdot)$ as decision (prediction) function, so task $\mathcal{T}$ can be expressed by $\mathcal{T} = \{y, f(x)\}$, where the decision function $f(x)$ predicts label y , for instance x .

Transfer Learning: Enhancing the performance of the target prediction function $f(x)_t$ in the target domain $\mathcal{D}_t$ through the use of knowledge gained from the learning task $\mathcal{T}_s$ and the source domain $\mathcal{D}_s$ is the aim of transfer learning. This can be achieved by adapting the knowledge from $\mathcal{D}_s$ and $\mathcal{T}_s$ to $\mathcal{D}_t$, enabling $f(x)_t$ to achieve improved accuracy or generalization. (It may be $\mathcal{D}_s \neq \mathcal{D}_t$ or $\mathcal{T}_s \neq \mathcal{T}_t$).

Given condition $\mathcal{D}_s \neq \mathcal{D}_t$, it follows that the feature spaces in both domains are not equivalent, leading to $\mathcal{X}_s \neq \mathcal{X}_t$ or a difference in the marginal distribution of instances across domains, as represented by $P(X)_s \neq P(X)_t$. Similarly, the condition $\mathcal{T}_s \neq \mathcal{T}_t$ indicates that the label spaces in both domains are not the same, resulting in $y_s \neq y_t$ or a divergence in prediction functions across domains, represented by $f_t(x) \neq f_s(x)$. This distinction highlights that the label set or conditional distribution differs when two tasks are dissimilar.

In single-source transfer learning, knowledge is transferred from one source domain, $\mathcal{D}_s$ to a target domain $\mathcal{D}_t$. The goal is to improve the target task $\mathcal{T}_t$ by leveraging knowledge from the single-source task $\mathcal{T}_s$. The challenge in this approach is that the source domain may be of limited relevance to the target domain, particularly when there are significant differences in the feature space or distribution of data. Despite these challenges, single-source transfer learning remains one of the most widely used approaches due to its simplicity and effectiveness in cases where the source domain is closely related to the target domain.

In multi-source transfer learning, knowledge is transferred from multiple source domains $\{\mathcal{D}_{s1}, \mathcal{D}_{s2}, \dots, \mathcal{D}_{sn}\}$ to a single-target domain $\mathcal{D}_t$. This method can provide more robust learning, allowing the model to incorporate diverse knowledge from multiple domains. However, combining information from various sources introduces new challenges, such as aligning feature spaces and managing differences in data distributions. Moreover, multi-source learning often involves incomplete or partial data from each source, making it crucial to develop methodologies that can handle such complexities (Wang et al. 2024).

Fuzzy systems are increasingly integrated into transfer learning to manage uncertainty, especially in cases where the data or the domain relationships are ambiguous or imprecise. Fuzzy logic provides a framework to model uncertainty and vagueness, offering flexibility in decision-making processes.

In transfer learning, fuzzy systems can be applied to the classification tasks within both source and target domains, where exact domain boundaries are unclear or where there is overlap in feature spaces. By incorporating fuzzy rules, transfer learning approaches can enhance adaptability and handle noisy or incomplete data more effectively.

Incomplete transfer learning refers to scenarios where complete data is lacking in the source domains. This situation is common in multi-source transfer learning, where some source domains may not have full data representations or sufficient labeled instances. In such cases, the challenge lies in transferring knowledge effectively despite these gaps. Approaches to handle incomplete transfer learning often involve developing algorithms that can learn from partial information or prioritize the most relevant and complete data sources while mitigating the impact of missing information (Li et al. 2024).

3.1 Methodologies and Categories

Transfer learning has emerged as a crucial technique in machine learning, particularly in scenarios where labeled data is limited in the target domain but plentiful in the source domain. This approach facilitates knowledge transfer from one domain (the source) to enhance learning performance in another (the target) domain. Transfer learning methodologies can be broadly classified based on the nature of the tasks and data involved. The primary categories include transfer learning based on label spaces, transfer learning based on feature spaces, and transfer learning based on the number of domains. This categorization is commonly found in other review articles; however, we have introduced an additional subcategory to enhance the framework. Among these categories, we will discuss applying fuzzy methods to address various challenges, particularly incomplete domains. By examining these aspects, we aim to comprehensively understand how transfer learning can be effectively implemented across diverse scenarios.

3.2 Transfer Learning Based on Label Access

Transfer learning can be classified into three main types based on the availability of labeled data: inductive, transductive, and unsupervised transfer learning. Each type leverages labeled data differently, offering unique strategies to address various tasks.

Inductive Transfer Learning: This approach assumes that labeled data is available in both the source and target domains. A model trained on the labeled source domain can be fine-tuned using labeled examples from the target domain to improve its performance on the target task (Chattopadhyay et al. 2012; Chen et al. 2019). Inductive transfer learning is

particularly effective when the source and target domains share similarities, as it allows the model to transfer learned features efficiently to the target task. This method is widely used in applications with labeled data in both domains.

Transductive Transfer Learning: In this scenario, labeled data is only available in the source domain, while the target domain contains only unlabeled data. The model leverages knowledge from the source domain to make predictions on the unlabeled target domain. Techniques like self-training are often employed, where the model generates pseudo-labels for the target domain data and iteratively improves its predictions. Transductive transfer learning is ideal for tasks where obtaining labeled data in the target domain is challenging or infeasible.

Unsupervised Transfer Learning: This type addresses situations where both the source and target domains lack labeled data. The model learns meaningful representations directly from the available unlabeled data without explicit supervision. Unsupervised transfer learning is particularly valuable in tasks where labeled data is scarce or nonexistent (Wilson and Cook 2020).

By understanding these three types of transfer learning, practitioners can select the most suitable approach based on the nature of their data and the task at hand. Each method provides distinct advantages, making transfer learning a powerful and flexible tool in modern machine learning.

4 Inductive Transfer Learning

Transfer learning involving label information between domains is called inductive transfer learning. In this approach, the tasks across domains are different ($T_s \neq T_t$), while the domains may or may not be the same ($D_s \neq D_t$ or $D_s = D_t$). The primary objective of inductive transfer learning algorithms is to enhance the learning of the objective function (Deng et al. 2014b). The conditions of these settings are established when well-labeled data is usually accessible in the target domain, regardless of whether sufficient labeled data is accessible or inaccessible in the source domain. The former is given particular attention. Commonly, the main goal of implementing inductive transfer learning is to develop the target model when a small set of target data is well labeled.

In this context, inductive learning shares similarities with multi-task learning, albeit with a distinct emphasis solely on the target task. It can also be likened to self-taught learning (Raina et al. 2007), especially when the source domain lacks labeled data. Since the data remains concealed within the source domain, direct transfer learning is impractical for its retrieval.

Inductive transfer learning involves applying knowledge from a different task (source task) to a new task (target task),

requiring some labeled data from the target domain to guide the transfer process, even if the domains are similar. Inductive transfer learning is one of the sub-categories of transfer learning, which is gaining popularity in training models based on using knowledge from the source domain for predicting the target domain.

Four types of solutions exist for solving the problems in inductive transfer learning situations: instances-based learning, feature-based learning, parameters-based learning, feature-based learning, and relationships-based learning. Furthermore, inductive learning is dominated by the first three categories of procedures, whereas relation-based approaches are uncommon.

TrAdaBoost is a prominent technique among instance-based inductive transfer learning algorithms, drawing its inspiration from the AdaBoost transfer learning algorithm outlined in Dai et al. (2007) and (Schapire 2009). TrAdaBoost employs iterative reweighting of source domain instances to extract valuable information from them. Initially, it assesses the significance of each old dataset in the source domain by utilizing a limited number of labeled new data, referred to as same-distribution data ($\mathcal{T}_s$). Additionally, it categorizes low-frequency occurrences as diff-distribution data ($\mathcal{T}_d$). To construct a new model for the intended task, TrAdaBoost combines $\mathcal{T}_d$, $\mathcal{T}_s$, and unlabeled data from $\mathcal{S}$. Notably, the reweighting process in TrAdaBoost differs from that of AdaBoost. It reduces the weights of instances in $\mathcal{T}_s$ that were correctly predicted while elevating the importance of examples in $\mathcal{T}_s$ that were inaccurately classified.

Furthermore, Yao and Doretto (2010) proposed an instance-based transfer learning method to mitigate negative transfer from unrelated source domains by leveraging knowledge from multi-sourcedomains through a boosting approach. Similarly, the study introduced a new multi-sourceboosting technique called GapBoost for transfer learning (Wang et al. 2019b). Some research endeavours focus on reducing marginal distribution disparities. One such approach is Transfer Component Analysis (TCA), as suggested by Che et al. (2021). TCA enforces the scatter matrix as a constraint and employs Maximum Mean Discrepancy (MMD) to quantify the differences in marginal distributions. TCA is a comprehensive method that involves learning linear mapping from an empirical kernel feature space to a low-dimensional feature space. The use of eigen-decomposition simplifies the solution to the optimization problem. In the realm of transfer learning, MMD is frequently employed, with its definition as follows:

$$MMD(X^s, X^T) = \frac{1}{n_s} \sum_{i=1}^{n_s} \Phi(x_i^s) - \frac{1}{n_T} \sum_{i=1}^{n_T} \Phi(x_i^T)^2_{\mathcal{H}} \quad (1)$$

One efficient way to calculate the MMD is through a kernel-based approach. This technique works by projecting the

data points into a more expansive space known as a Reproducing Kernel Hilbert Space, or RKHS and then measuring the distance between the average representations of the two distributions in this space. This distance reflects the overall difference between the two distributions. In another way, the kernel approach offers a computationally efficient way to calculate the MMD, making it a valuable tool for distribution comparison tasks.

The marginal distribution can be reduced by optimizing an objective function that simultaneously performs source domain sampling selection and marginal distribution adaptation. This process aligns the distributions of the source and target domains, improving the effectiveness of transfer learning (Long et al. 2014). Before doing the feature map, Distribution Matching Embedding (DME) (Baktashmotlagh et al. 2016) computed the transformation matrix. The global and local marginal distributions were updated using a different ITCA technique (Jiang et al. 2017).

Zellinger et al. (Zellinger et al. 2017) achieved higher-order moment matching (CMD) between domain distributions by leveraging the property of MMD as an implicit moment matching procedure, similar to the concept of central moment discrepancy proposed in Baktashmotlagh et al. (2016). As an empirical estimate, the domain discrepancy in the activation space $[a, b]^N$ is provided by the following formula:

$$CMD_K(X^s, X^T) = \frac{1}{b-a} \frac{1}{n_s} \sum_{i=1}^{n_s} (x_i^s) - \frac{1}{n_T} \sum_{i=1}^{n_T} (x_i^T)^2_{\mathcal{H}} + \sum_{k=1}^K \frac{1}{(b-a)^k} C_K(X^s) - C_K(X^T)^2_{\mathcal{H}} \quad (2)$$

where $C_K(X^s) = E(X - E(X))^k$ represent the vector of all k^{th} order sample central moments. However, some TrAdaBoost-inspired algorithms have achieved improved performance. Notably, one of TrAdaBoost's limitations is that it relies on a single type of base learner to train the target domain model, even though alternative base learners may exist and offer superior performance. While several approaches like (Matasci et al. 2012) and (Yao and Doretto, 2010) exist for leveraging diverse base learners to enhance task-specific performance, the original TrAdaBoost algorithm solely utilizes knowledge from a single-source domain, which can be insufficient in certain situations. Papers (Matasci et al. 2012) and (Yao and Doretto 2010) take advantage of integrating many source datasets to minimize negative learning to get around this weakness. The method in Eaton and Desjardins (2011) intelligently selects the most relevant sources through a two-step boosting process, prioritizing specific instances and refining them based on task-specific needs. In addition to task-based boosting, which can improve the examples from

tasks with higher transferability, it also boosts the weights of mistakenly predicted instances.

In contrast to TrAdaBoost, some methods maintain all primary learners, which can enhance model performance by allowing earlier iterations to focus on the core aspects of the data while later iterations refine more intricate patterns. Additionally, certain studies have applied dynamic weight update techniques to further improve model performance. By iteratively re-weighting instances, these approaches have been shown to enhance the effectiveness of inductive learning models. However, this approach is not without controversy. Some researchers argue that diverse data portions from the source domain can have mixed effects on model performance in the target domain. While certain data subsets can aid the model by providing beneficial information, other portions may be detrimental, potentially introducing noise or biases that misguide the model. Furthermore, accurately measuring the transferability of source data to a target domain remains a complex task, as there is no simple way to evaluate which parts of the source data will contribute positively to model performance. This uncertainty can challenge the effectiveness of re-weighting strategies in transfer learning.

To reduce domain divergence and model error, feature-based inductive transfer learning techniques (Long et al. 2017b; Yosinski et al. 2014), and (Afridi et al. 2018) frequently seek to extract shared features. Feature-based domain adaptation algorithms can be categorized into two main groups: supervised and unsupervised, depending on the availability of labeled data in the source and target domains. Supervised Feature-Based Algorithms require substantial labeled data in both the source and target domains. They typically employ multi-task learning, where a single model learns to perform multiple tasks simultaneously, one for each domain. This approach leverages the labeled data from both domains to improve the model's generalization ability in the target domain. To create a high-performance model for the target domain, even with limited labeled target data, algorithms like those in Eaton and Desjardins (2011); Long et al. 2017b), and (Yosinski et al. 2014) leverage a wealth of labeled source data alongside the available target data. Unsupervised Feature-Based Algorithms are designed for cases with only available labeled data in the source domain. They aim to extract domain-invariant features by minimizing the discrepancy between the source and target domain data distributions. This allows the model to learn a representation independent of the domain shift and can be effectively applied to the target domain. Common techniques in unsupervised feature-based algorithms include adversarial training, MMD minimization, and domain-regularization methods. These approaches encourage the model to learn features that are relevant to the task while minimizing the domain-specific information. Inductive transfer learning,

on the other hand, concentrates on the target task, whereas multi-task learning prefers to learn all the tasks at once. On the other hand, unsupervised algorithms are more robust yet more challenging to train (Salimans et al. 2016).

Geodesic Flow Kernel (GFK), proposed by Gong et al. (Gong et al. 2012), is a subspace feature alignment method designed to address the challenges in domain adaptation. GFK is an extension of the earlier Geodesic Flow Subspaces (GFS) method, which leverages incremental learning principles and aims to exploit the knowledge encoded within the geodesic path between two domains. Below is an overview of the GFS framework, which serves as the foundation for GFK:

The first step in the GFS framework involves the generation of subspaces for the source and target domains. This is achieved independently through PCA, where both domains are projected onto their respective subspaces, forming the basis for the following alignment stages.

Once the subspaces are established, GFS treats them as discrete points on the Grassmann manifold. By analyzing the geometric properties of this manifold, GFS identifies a discrete set of intermediate subspaces along the geodesic path between the source and target subspaces. These subspaces act as bridges that reduce the domain discrepancy in the feature space.

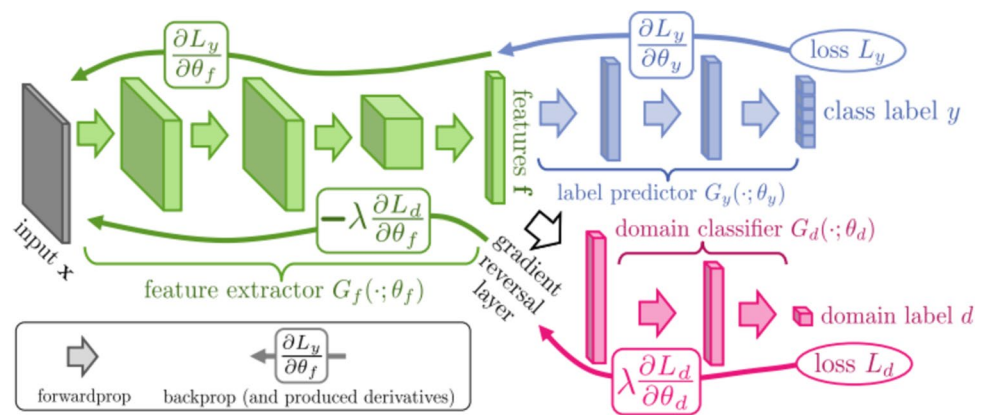
In the final stage, feature vectors from both the source and target domains are projected onto their respective subspaces. These projected features, including those from the interpolated subspaces, are stacked together to form a new feature representation. This method allows for better alignment of features across domains.

A key limitation of the GFS approach is determining the appropriate number of interpolated subspaces. This selection process can be computationally demanding and suboptimal if not properly tuned. To address this, GFK extends GFS by integrating an infinite number of subspaces along the geodesic path between the source and target domains, thereby overcoming the need for manual subspace selection.

The primary innovation of GFK lies in its construction of a high-dimensional feature space that incorporates information from all subspaces along the geodesic flow. This is achieved by defining a kernel function that efficiently computes inner products in the resulting infinite-dimensional space, enabling better feature alignment without explicitly needing to compute each intermediate subspace.

Additionally, GFK introduces two important measures to enhance its applicability: This metric helps determine the optimal subspace dimensionality for feature alignment, ensuring that the model captures the most relevant information from both domains. In scenarios involving multiple source domains, this metric identifies the most suitable source domain by evaluating the relevance of each source to the target domain.

Fig. 6 The framework of DANN (Ganin et al. 2017)



By addressing the limitations of GFS and introducing these novel metrics, GFK represents a significant advancement in subspace-based domain adaptation methods.

Recent research, including studies by Long et al. 2017b; Yosinski et al. 2014; Salimans et al. 2016), and (Ganin et al. 2017), has investigated the integration of transfer learning and Generative Adversarial Networks (GANs). This approach aims to eliminate the need for human involvement in the manual construction of networks for extracting shared features. Instead, it leverages the capabilities of GANs to automatically discover and extract these shared properties, enabling the utilization of transfer learning for various tasks. In (Ganin et al. 2017), the authors leverage the powerful concept of Generative Adversarial Networks to identify features that are both crucial for the primary learning task in the source domain and remain invariant across domains. This innovative approach, known as the Domain-adversarial Neural Network (DANN) and illustrated in Fig. 6, offers significant advantages for transfer learning.

DANN employs a traditional feed-forward neural network architecture, seamlessly combining a deep feature extractor and a deep label predictor. This structure enables the model to extract meaningful features from input data and generate accurate predictions based on those features. The mechanism for unsupervised domain adaptation involves the incorporation of a domain classifier, which is connected to the feature extractor. This connection is facilitated through a gradient inversion layer, wherein a specified negative constant inverts the gradients during backpropagation-based training. Otherwise, training continues as usual for source samples, reducing domain classification loss and, for all instances, label prediction loss. Gradient inversion guarantees that the feature distributions in the two domains are comparable (as close to being undetectable to the domain classifier as possible), leading to domain invariant features. They define the Loss function as follows:

$$E(\theta_f, \theta_y, \theta_d) = \sum_{i=1, d_i=0}^N L_y^i(\theta_f, \theta_y) - \lambda \sum_{i=1}^N L_y^i(\theta_f, \theta_d) \quad (3)$$

where L_d is the loss for domain classification and L_y is the loss for label prediction.

Also, Some simple regularization strategies using convolutional networks have been developed in the paper (Li et al. 2020a), namely L2. They have a clear bias toward the solution learned on the source task, resulting in a trade-off between the target task solution and the pre-trained parameter consistent with the goal for fine-tuning. Another group of methods has used fuzzy systems and deep networks in transfer learning problems.

Generalized Hidden-Mapping Ridge Regression (GHRR), introduced by Deng et al. 2014b, is a versatile technique for training various classical intelligence models, encompassing neural networks, kernel methods, and fuzzy logic systems. GHRR incorporates a knowledge-leveraging transfer learning mechanism to facilitate inductive transfer learning, resulting in the Transfer GHRR (TGHRR) approach. TGHRR leverages induced knowledge. This knowledge is more concise and clearer compared to information extracted directly from source domain data, leading to improved control and achieving optimal alignment between source and target data manifolds. The quantity of information provided and the level of confidence in mastering a task are correlated; tasks with limited information have a high degree of uncertainty. Extracting a finite amount of information from insufficient target domain data with answers results in high uncertainty.

Nevertheless, fuzzy systems have shown promise in resolving this issue. The framework structure contains the fuzzy transfer learning approach. Figure 7 demonstrates the essential elements. The idea of using the fuzzy system to transfer and apply their relations and customise fuzzy components using relevant information is clear in this framework (Shell and Coupland 2015). Some studies, like (Deng et al. 2013a), also use the fuzzy system in inductive

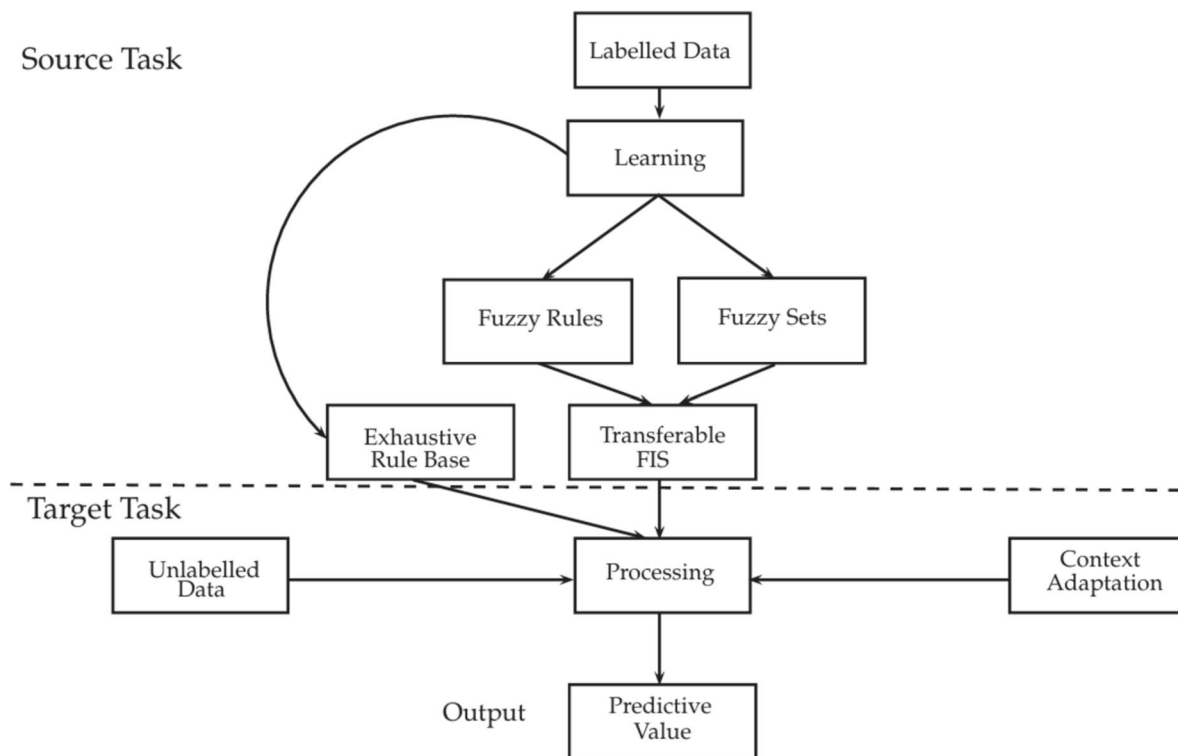


Fig. 7 Fuzzy Transfer Learning framework (Shell and Coupland 2015)

transfer learning. For instance, the study's authors (Deng et al. 2016) developed the Takagi Sugeno Kang fuzzy system model based on increased knowledge leverage. The proposed method incorporates two distinct knowledge-leveraging strategies within an improved algorithm designed to enhance parameter-based learning of the Takagi Sugeno Kang fuzzy system model in the target domain by effectively utilizing knowledge from the source domain.

One method is applied to learn antecedent parameters, while the other is used to understand the following parameters. The presented model, EKL-TSK-FS, has better transfer learning capacities than the other proposed methods. Also, the design model can provide acceptable performance in multi-source environments. The proposed algorithm does not improve negative transfer learning sufficiently; furthermore, the research authors (Han et al. 2021) built and analyzed interactive transfer learning as an inductive transfer learning method. It can reduce negative transfer between domains to improve fuzzy neural network learning performance, ITL-FNN. At first, an information filter method is created to rebuild the information in the original scene while balancing matching accuracy and diversity. The knowledge from the source domain can then be used to fit the instance of the target domain accurately.

Furthermore, a self-balancing technique is used to equalize the driving information between the source and target

domains. The knowledge can then be reorganized to eliminate redundant data. Finally, a structural competition method is established to adapt the fuzzy neural network knowledge.

5 Transductive Transfer Learning

Transductive learning becomes applicable when dealing with identical tasks but potentially disparate domains. In this scenario, access to labeled data is confined exclusively to the source domain. Learning on instances and learning on features are the two different learning methods that make up transductive transfer learning. The most well-known instance of transductive learning is domain adaptation. Transfer learning should first predict the distribution of labels on the target domain using the source domain data. Due to the drawbacks of its learning mechanism, instance-based approaches in transductive transfer learning are less common than feature-based methods. Therefore, feature-based techniques are the current mainstream transductive transfer learning (Gui et al. 2018).

Algorithms for learning from instances are primarily characterized as methods for transferring information from one domain to another by resampling or reweighting source examples. Additionally, instance-based approaches are predicated on two rigid premises: Under the assumption

of equivalent conditional distributions in both domains, the approach leverages a subset of source domain training data, establishing specified linkages with the target domain for effective model adaptation. Target domain training often necessitates judiciously selecting samples from the available source data, as not all of it may prove equally pertinent. This underscores the importance of carefully curating informative instances that best bridge the gap between the source and target domains. Recent advancements in domain adaptation techniques offer valuable tools for achieving this. For instance, researchers in paper (Yao and Doretto 2010) introduced a boosting algorithm that explicitly incorporates domain discrepancy measures into the selection process, prioritizing samples that minimize the domain shift while maximizing data informativeness.

Furthermore, alternative instance-based adaptation models, as referenced in Xu et al. (2018) and (Xia et al. 2014), demonstrate the capability to address tasks characterized by high-dimensional distributions. The underlying concept in these models involves employing a logistic approximation to bridge the gap between data from the source and target domains. More recently, a new approach was introduced in Lee et al. (2019) and further explored in Bell (1962), leveraging maximum correlation analysis for instance-based multi-source transfer learning. A noteworthy feature of this approach is that it eliminates the need to train a target domain model directly on source domain data. Instead, it relies on pre-trained models from the source domains to create a network of interconnected feature extractors for the target domain data.

In an effort to enhance transfer learning performance, a novel maximum correlation metric (Wang et al. 2019b) was introduced to measure the separation between source and target distributions. This metric, designed to handle domain discrepancies, plays a key role in bridging the gap between different domains in the transfer learning process. Additionally, the study proposed four guidelines tailored to the development of transfer learning algorithms, which are essential for achieving effective adaptation across domains. These guidelines are as follows: The first principle is minimising the weighted empirical loss over source and target domains. This ensures that the model learns from the combined data effectively, reducing the divergence between the two distributions. The second guideline recommends assigning a balanced weight to each data point to counter the overfitting that can arise from perturbations in training data. This balanced weighting scheme is crucial for maintaining the robustness of the model, particularly in the presence of noisy or incomplete data. Since the target data will ultimately be used for testing, the third principle advocates for assigning more weight to the target samples during training. This

approach ensures that the model focuses more on the domain of interest, improving generalization performance. The final guideline suggests minimizing the performance disparity between the source and target domains by ensuring that the assigned weights do not overly favor one domain. This balance is critical for achieving consistent performance across both domains.

In addition to these guidelines, the research introduced GapBoost, a novel method that adjusts the instance weight matrix based on a new domain distance measurement, referred to as *Y-Discrepancy*. The *Y-Discrepancy* is defined as:

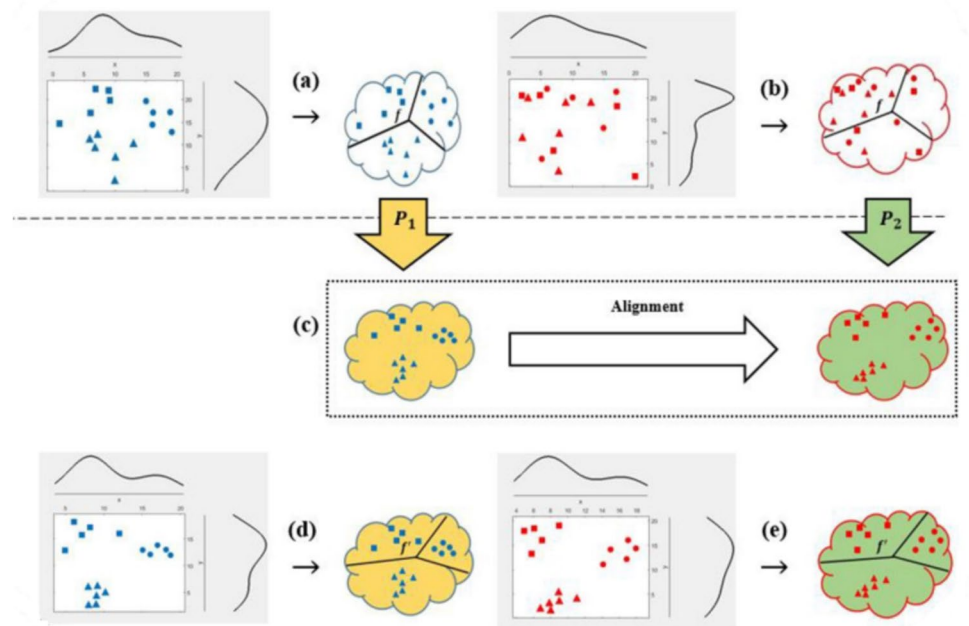
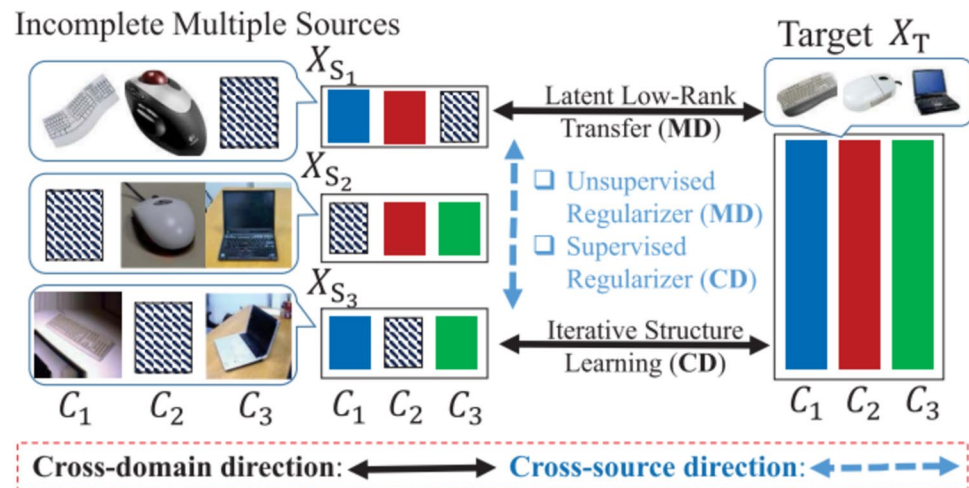
$$dist_Y(\mathcal{D}_s, \mathcal{D}_t) = \sup |L_{\mathcal{D}_s}(h) - L_{\mathcal{D}_t}(h)|, h \in H \quad (4)$$

Here, h represents the hypothesis, or the selected ideal learning model, and $dist_Y$ quantifies the maximum difference between the loss functions over the source ($\mathcal{D}_s$) and target ($\mathcal{D}_t$) domains for any hypothesis $h \in H$.

By iteratively adjusting the weight matrix using *Y-Discrepancy* during each training iteration, GapBoost dynamically aligns the source and target domains, ensuring that the model performs optimally. This method significantly reduces computational complexity during training, making it a promising technique for efficient and scalable domain adaptation.

However, many real-world issues (Sun and Saenko 2016; Yan et al. 2017), and (Salimans et al. 2016) do not necessarily meet these instance-based algorithms' prerequisites. As an alternative, techniques based on features have been created to address the problems. The concept of transferable characteristics for deep neural networks was first described by Fernando et al. (2013). The assumption for learning on features is often simply weaker: the source and target domain distributions are comparable. Moving or representing features to another space naturally tries to reduce the distribution mismatch between the source domain and the destination domain. Data-centred approaches (Sun and Saenko 2016; Long et al. 2019) and subspace-centred methods (Yan et al. 2017; Long et al. 2016a) are the two main categories of feature-based transductive learning methods.

Rezaei et al. (2021) present an innovative transductive transfer learning strategy for detecting unique and common features in source and target domains, namely TTLIC. The proposed learning approach maps both domains into subspaces with the least possible marginal and conditional distribution divergences. Racist and discriminatory learning across domains has been shown to increase model performance. Consequently, the proposed method differentiates domain classes by widening the gap between samples with different labels while narrowing the gap between instances

Fig. 8 The framework of TTLC (Rezaei et al. 2021)**Fig. 9** The architecture of TL-IMS (Ding et al. 2018)

sharing the same classes. The graphical representation of this method's framework can be observed in Fig. 8.

In addition to many advantages, the proposed algorithm in c does not provide acceptable performance in multi-source environments.

Handling incomplete or partial data across multiple sources in multi-source domain adaptation is especially challenging when aiming to enhance knowledge transfer for better learning in the target domain. The study presented in Ding et al. (2018) addresses this problem by proposing a unique transductive transfer learning framework designed to capitalize on knowledge from multiple, incomplete sources. This framework is ideal for scenarios where each source domain lacks complete data coverage, which poses a barrier

to traditional knowledge transfer approaches. Figure 9 in the study provides a schematic representation of incomplete sources, highlighting how this framework addresses data gaps across sources.

To overcome these limitations, the authors present multi-source transfer learning as an incomplete learning method, using two primary directions of knowledge transfer: cross-source and cross-domain transfer.

Cross-source transfer enables the exchange of information among multiple source domains, fostering interaction and knowledge sharing between incomplete or partially overlapping datasets. This collaborative exchange helps to bridge informational gaps, enhancing the collective representation of knowledge available to the model. Conversely,

cross-domain transfer focuses on aligning the aggregated multi-source knowledge with the target domain, ensuring that the transferred information is effectively adapted and optimized for the target task, even in the presence of incomplete or fragmented data.

In the schematic depicted in Fig. 9, each color corresponds to a distinct class, illustrating a scenario where none of the target labels X_T are fully represented by any individual source domain X_{S1} , X_{S2} , or X_{S3} . This highlights the complexities introduced by incomplete source domains. To overcome these challenges, the proposed framework integrates iterative structure learning and latent low-rank subspace transfer to establish direct connections between the source and target domains, effectively minimizing distributional discrepancies. Furthermore, two cross-source alignment mechanisms act as bridges between the source domains, promoting information exchange and addressing the absence of class labels across domains. This dual-directional transfer strategy allows the model to harness knowledge from fragmented and incomplete multi-source datasets, enabling robust and effective learning in the target domain. This approach mitigates the limitations of incomplete sources and enhances the model's adaptability to complex, multi-domain environments.

Some papers use the fuzzy system to transfer knowledge in transductive transfer learning, like inductive transfer learning (Behbood et al. 2015, 2014). Bank failure prediction models with domain-specific data distributions for long-term forecasting have been introduced by Behbood et al. (Behbood et al. 2015, 2014) using a fuzzy-based transductive transfer learning technique. They used three common predictors to predict the initialize labels for data in the target domain: neural network, fuzzy neural network, and support vector machine. Then, they improved the labels using fuzzy similarity metrics. Another application of the transductive transfer learning system proposed by Xie et al. (Xie et al. 2019a) uses a fuzzy system. For identifying electroencephalogram signals, Xie et al. (Xie et al. 2019a) offered a transductive-based transfer learning system. The authors propose a training technique model based on a transductive learning algorithm that might be utilized more broadly. Firstly, the representation of several traditional intelligent models, such as fuzzy systems, feed-forward neural networks, and kernel regression models, is integrated using a generalized hidden-mapping model (GHMM). The GHMM is then supplied with an approach for obtaining transductive transfer learning. As a consequence, the recommended method is successful in training many classical intelligent models, and transductive learning can increase the generalization skills of the taught model for beneficial applications in the real world, like EEG signal identification, where data isn't sufficient for model training.

6 Unsupervised Transfer Learnin with Unlabeled Data

While inductive transfer learning tasks typically involve labeled data in both the source and target domains, unsupervised transfer learning operates in scenarios where neither domain has labeled data. In the absence of explicit labels, these algorithms leverage the inherent relationships and correlations within the data to learn useful representations. Rather than depending on labeled samples, the focus shifts to uncovering meaningful patterns within both domains using techniques such as dimensionality reduction, feature extraction, and clustering.

In unsupervised transfer learning, algorithms attempt to exploit the natural similarities between the source and target domains. This can involve analyzing the similarity of data points across domains, identifying shared features, or discovering common statistical properties. The learned representations and relationships are then used to apply the knowledge from the source domain to the target domain, enabling the algorithm to make predictions or perform tasks without explicit labels.

Various approaches for clustering and dimensionality reduction have been proposed to facilitate learning in feature space. For example, Self-taught Clustering (STC), as presented by Dai et al. (2008), allows for effective clustering in scenarios with limited labeled data in the target domain by using abundant unlabeled data from the source domain. STC effectively combines data from multiple domains into a unified feature space, enabling clustering the target data. Similarly, Transferred Discriminative Analysis (TDA) (Wang et al. 2008) uses clustering techniques to generate pseudo-class labels for target data.

Another notable approach is the Self-taught Learning algorithm (Raina et al. 2007), which uses unlabeled data to generate higher-level features through sparse coding, significantly enhancing performance in classification tasks.

Liu et al. (Liu and Qin 2020) introduced a novel domain adaptation method for unsupervised transfer learning called MKJDM. This method utilizes multiple kernels, unlike traditional single-kernel approaches, to evaluate distribution distances. The multi-kernel joint domain matching strategy captures a broader range of distribution discrepancies between the source and target domains. While single-kernel methods minimize the MMD metric under a single kernel function, the MK-MMD approach employs multiple kernel functions, providing a more comprehensive evaluation of discrepancies. MKJDM further improves performance by aligning both marginal and class-conditional distributions and employing instance-specific reweighting.

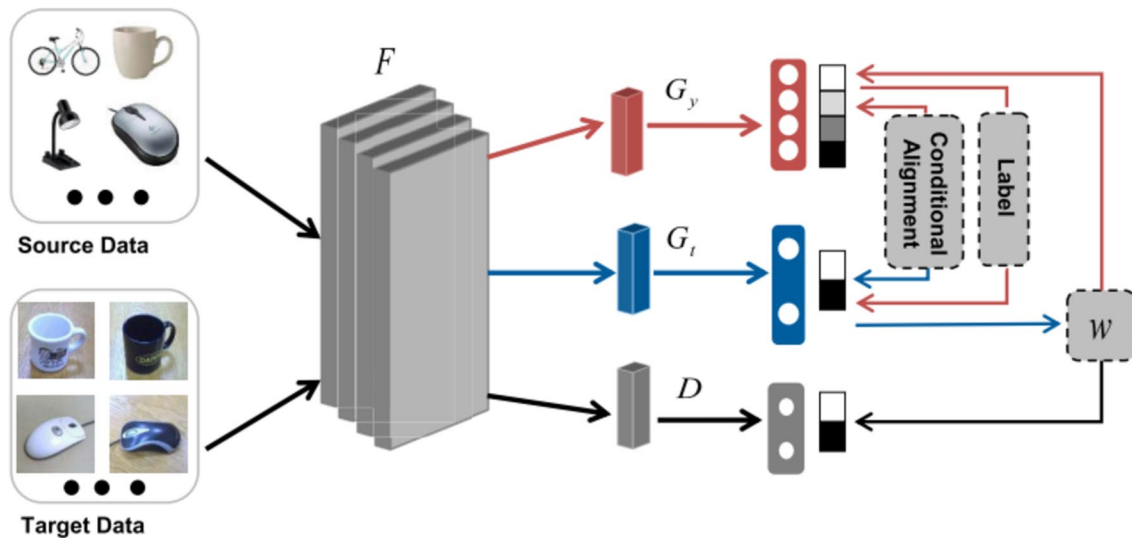


Fig. 10 The architecture of CLA (Yang et al. 2022)

While deep neural networks have been effective in various domains (Sermanet et al. 2014; Lecun and Zeiler 2014), their training requires large quantities of labeled data, which are often unavailable. To address this limitation, (Chang et al. 2018) proposed a new method for unsupervised transfer learning based on multidisciplinary convolutional sparse coding (MSCSC). MSCSC enables the learning of filters at different scales and performs tasks in the target domain without supervision. This method was successfully applied to biomedical datasets, demonstrating its effectiveness in achieving accurate results. However, one limitation of methods like paper (Chang et al. 2018; Chen et al. 2022b), and (Liang et al. 2020) is that they do not tackle the critical issue of selecting appropriate samples for feature transfer.

Yang et al. (Yang et al. 2022) proposed a method using deep neural networks with a contrastive learning approach and a conditional alignment technique. This method assigns more weight to related samples while reducing the influence of outliers. The conditional alignment (CLA) is supported by contrastive learning, which enhances the alignment between the source and target domains. Furthermore, Generative Adversarial Networks (GANs) are used in adversarial domain learning to bridge the gap between two domains, enabling a classifier trained on one domain to be applied to the other. The proposed architecture consists of a feature extractor F , a domain discriminator D , a classifier G_y that learns domain-invariant structures, and a classifier G_t that clusters the target domain data (see Fig. 10).

Continuing from the discussion on unsupervised transfer learning methods, Unsupervised Open-Set Domain Adaptation (UOSDA) introduces an additional layer of complexity due to the uncertainty in the target domain. While traditional unsupervised domain adaptation assumes that the target domain

contains only classes seen in the source domain, real-world scenarios often involve unknown classes in the target domain that are not present in the source domain. This discrepancy can lead to misclassifications and hinder the generalization of models trained on the source domain (Fang et al. 2021).

Unsupervised open-set domain adaptation aims to address these challenges by learning shared features between the source and target domains while simultaneously identifying and separating unknown classes. This is done through techniques such as anomaly detection, cluster analysis, and metric learning, which help distinguish known from unknown classes. By effectively handling unknown classes, unsupervised open-set domain adaptation methods have the potential to significantly improve the performance of models trained on unlabeled target data, especially in tasks like image classification, natural language processing, and medical diagnosis (Wang et al. 2023).

One of the key challenges in unsupervised open-set domain adaptation is accurate unknown class detection, which requires developing robust methods that can identify and separate unknown classes from the known classes in the target domain. Additionally, handling multiple unknown classes is another challenge that unsupervised open-set domain adaptation methods must address, ensuring that these unknown classes are not misclassified as known classes. Furthermore, there is a need for a theoretical understanding of unsupervised open-set domain adaptation to design more effective algorithms and establish performance bounds for these methods (Zhang et al. 2022a).

For instance, the work by Fang et al. (2021) provides a theoretical foundation for open-set domain adaptation by considering the risk associated with identifying unknown classes. Their approach introduces a learning bound that includes an "open

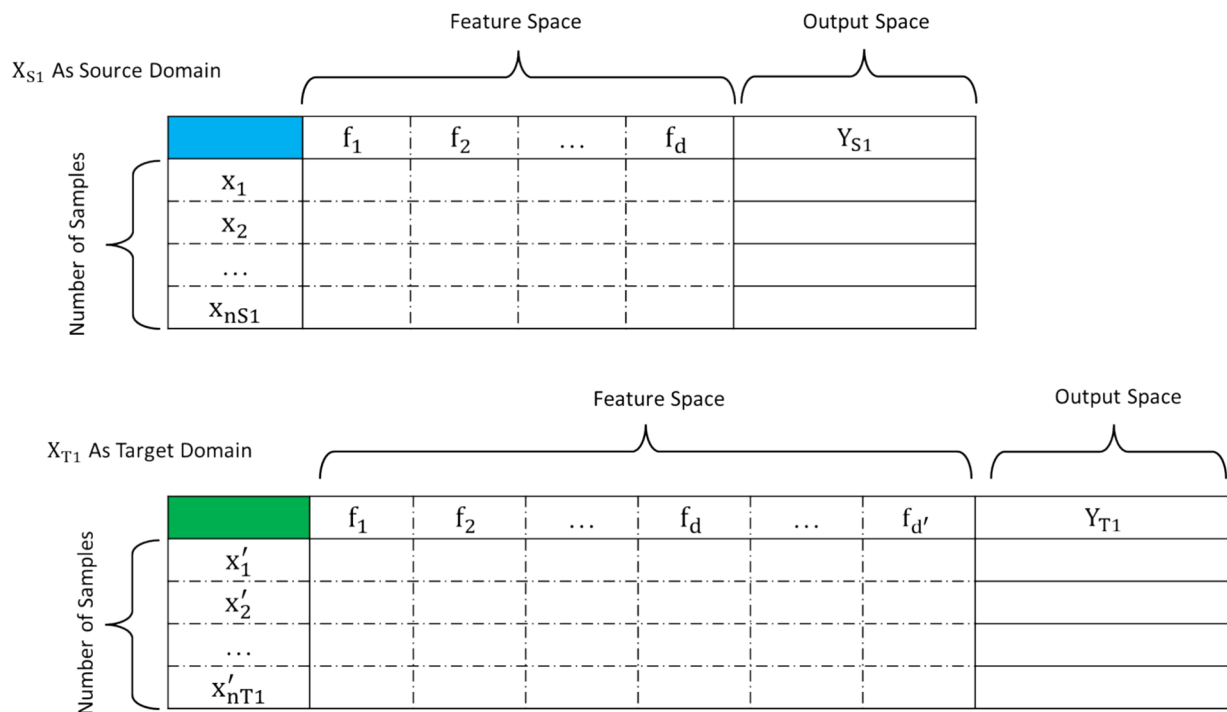


Fig. 11 Different settings between heterogeneous and homogenous conditions

set difference" term, reflecting the potential risk of misclassifying unknown classes. They also propose a novel unsupervised method called Distribution Alignment with Open Difference (DAOD), which regulates the discrepancy between open sets and enhances the adaptation process in open-set scenarios.

These advancements in unsupervised open-set domain adaptation represent a significant step forward in addressing the complexities of domain adaptation where unknown target classes are a concern, ultimately improving the generalization and robustness of models in real-world, unlabeled data environments.

7 Transfer Learning based on Feature Spaces

Magine transfer learning as a bridge between two knowledge islands: Homogeneous transfer learning: The bridges are perfectly aligned, allowing knowledge to flow smoothly between islands with matching feature dimensions. Heterogeneous transfer learning: The bridges need creative adjustments to accommodate differences in feature dimensions, ensuring knowledge can still cross despite the variations. For clarity, Fig. 11. shows the different settings between heterogeneous and homogenous conditions. In heterogeneous conditions, the feature spaces in domains are different, in contrast to homogeneous conditions, which are the same. X_{S1} is a source domain that has X_{nS1} samples with dimension f_d and X_{T1} is a source domain that has X_{nT1} samples

with dimensions $f_{d'}$. In heterogeneous conditions, $f_d \neq f_{d'}$ and in homogeneous conditions, $f_d = f_{d'}$. In this example, it is assumed that there is only one source domain to transfer knowledge to one target domain.

8 Homogeneous

The transfer of learning in a homogeneous environment defines a situation in which the feature space and dimensional of the source and target samples are the same, which means $\mathcal{X}_s = \mathcal{X}_t$ and $d_s = d_t$ which d shows the dimensional of feature space. However, the associated probability distribution, marginal probability distribution, is different $P(X)_s \neq P(X)_t$. The primary goal is to align feature distributions of two domains. There are various approaches to solving homogeneous transfer learning problems. Most homogeneous transfer learning strategies consider one of three general strategies: (1) modifying the marginal distribution difference in the source, (2) modifying the conditional distribution difference in the source, and (3) modifying both conditional and marginal distribution differences in the source.

These solutions are under two common frameworks, the subspace-based method and the deep feature learning method. They all have the same fundamental idea: find suitable feature representations by exploring latent common feature subspaces or extracting deep high-level features. Homogeneous transfer learning is divided into two

categories based on the presence of the label in the target domain: semi-supervised (Donahue et al. 2013) and unsupervised (Pan et al. 2011).

TCA (Pan et al. 2011) is one of the homogeneous domain adaptation methods where the target domain samples are unlabeled. Its purpose is to reduce the difference in the marginal distribution between domains, which maximizes their variance to preserve the original data's characteristics. Cat (Gong et al. 2014) is another domain adaptation method that reduces the marginal distribution between domains by putting the data on the manifold. SA (Fernando et al. 2013) is also one of the dimension reduction methods in matching the domain, which matches the rough spaces together. Furthermore, theoretical information learning (Shi and Sha 2012), transfer deep network (Long et al. 2016b), sampling geodesic flow (Gopalan et al. 2014), scatter component analysis (Ghifary et al. 2017) and feature-level domain adaptation (Kouw et al. 2016) are representative methods for homogeneous domain adaptation.

Proposing the Discriminative Transfer Joint Matching (DTJM) approach, the research (Peng et al. 2019) aims to bridge the gap between source and target features by minimizing empirical MMD in a kernel principal component analysis space, promoting alignment for effective transfer learning. Subsequently, it employs an $\ell_{2,1}$ norm on the embedding matrix, such as reweighting. This step is crucial for preserving the local manifold structure across diverse domains while simultaneously maximizing the dependency between the embedding and corresponding labels.

The study (Zhao et al. 2021) incorporates a unified framework that combines transfer learning and metric learning to mitigate the dissimilarity in distribution between the source and target domains. This integration aims to acquire more distinctive feature representations during knowledge transfer.

Furthermore, the paper introduces Geometric Mean Transfer Learning (GMTL) as an innovative and robust transfer learning technique grounded in geometric mean metric learning. GMTL models intra-class distance and inter-class similarity, employing weighted geometric mean metric learning. Notably, the existing models discussed do not evaluate the quality of the acquired features.

Zhang and Davison (2021) propose an initial approach called rapid subspace sampling demons (SSD) to learn intermediate subspace features from two domains swiftly. Subsequently, they evaluate the effectiveness of these learned features.

A new domain adaption approach that can comprehensively analyze the target domain's data distribution structure has been proposed by Tian et al. (2020). Their proposed framework is shown in Fig. 12. Rather than treating samples inside the same cluster in the target domain as individuals, their method treats them as a group and gives pseudo-labels to the target cluster using class centroid matching. The authors address the challenge of capturing complex local relationships within the target data by implementing a self-learning approach. This technique empowers the model to dynamically identify and utilize the inherent connections between neighbouring samples, enriching its understanding of the target domain's manifold structure. The objective function of their proposal is solved using an effective iterative optimization technique with a theoretical convergence guarantee. The proposed method extends beyond unsupervised domain adaptation to a semi-supervised scenario that includes homogeneous and heterogeneous conditions in a single domain. Instead of deep neural networks, some studies use the fuzzy system to overcome the problem in homogeneous spaces. Lu et al. (Lu et al. 2020b) have proposed a fuzzy logic method for homogeneous and heterogeneous environments. They

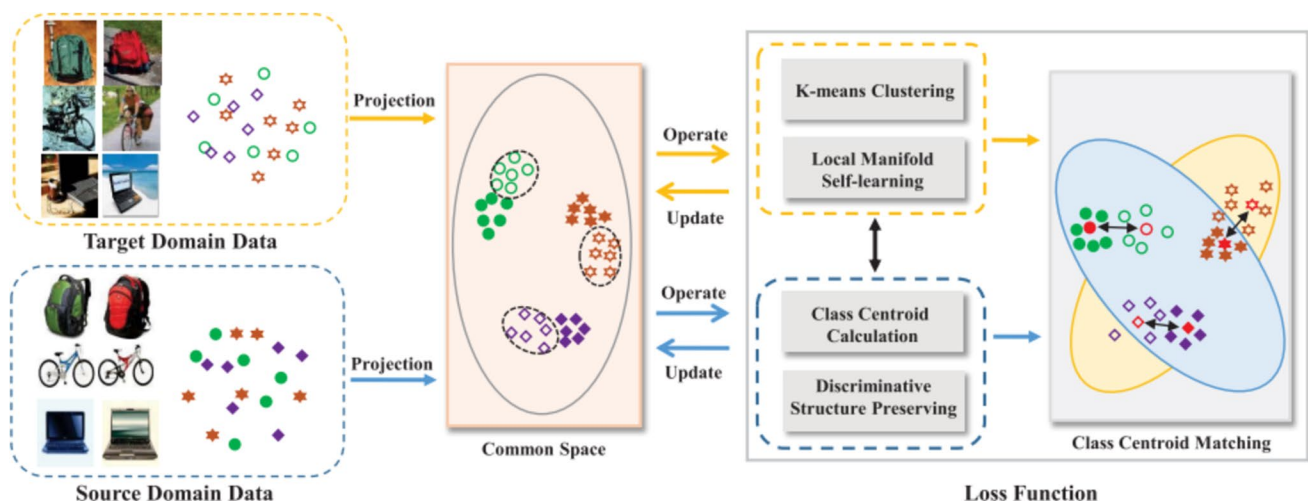


Fig. 12 The architecture of CMMS (Tian et al. 2020)

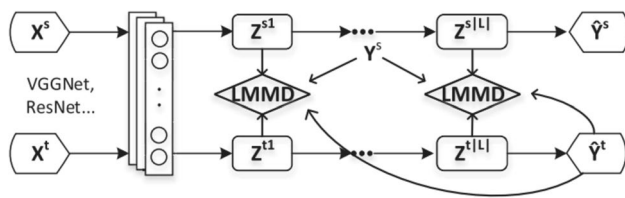


Fig. 13 Architecture of DSAN (Zhu et al. 2021)

believe that since there is very little information in transfer learning problems, there is a higher degree of uncertainty, so using a fuzzy system can report good results. The paper determines three main steps in homogeneous space: combining fuzzy rules extracted from source domains, selecting the best rules, and modifying rules to deal with the regression task's target domain with a few labeled data. Also, they accommodate the notion of heterogeneous space employing multi-source domains. The source domain data map to a latent feature space communal, which can transfer rules, and then they pass the steps of the homogeneous space procedures. The output results express the validity of their methods.

9 Heterogeneous

Building accurate models becomes tricky when working with diverse data sets, where one lacks the same features as the other. Heterogeneous transfer learning steps in to tackle this very challenge. This technique bridges the gap between domains with fundamentally different features, allowing knowledge gained from one to be effectively applied to the other, ultimately building a robust predictive model for the target domain despite its distinct feature set. Heterogeneous transfer learning methods may be developed in supervised, semi-supervised, or unsupervised situations (Li et al. 2014; Liu et al. 2021c). A primary model in supervised conditions is a heterogeneous spectral mapping model, such as a heterogeneous map, such as HeMap (Shi et al. 2013), which connects the target and source domains' unique feature spaces. Through spectral embedding, even when the feature spaces are entirely different (Shi et al. 2013).

The idea of optimum transport, a robust technique initially created for aligning two different distributions, has been used in Yan et al. (2018) by Yan et al. to provide a unique heterogeneous semi-supervised model. Although progressive alignment, as PA (Li et al. 2019), uses an unsupervised technique to learn representations of two heterogeneous domains, it still requires labeled instances to train a final classifier from the target domain that can overcome negative transfer learning.

In addressing the challenges of visual domain disparities, the research presented in Wang et al. 2020a introduces a

novel approach called Coupled Adversarial Transfer Domain Adaptation (CatDA). CatDA is designed as a shallow semi-supervised adversarial transfer learning network to align the distributions between two domains. Drawing inspiration from CycleGAN, CatDA diverges from deep networks and opts for multiple shallow multilayer perceptrons (MLPs). More specifically, the framework of coupled adversarial learning is employed in CatDA, as it comprises two symmetric and lean sub-networks. Input data from the source and target domains can be fed into the MLP network, generating the source and target domains under the guidance of two opposing linked discriminators, a configuration made possible by the symmetry of the two generators.

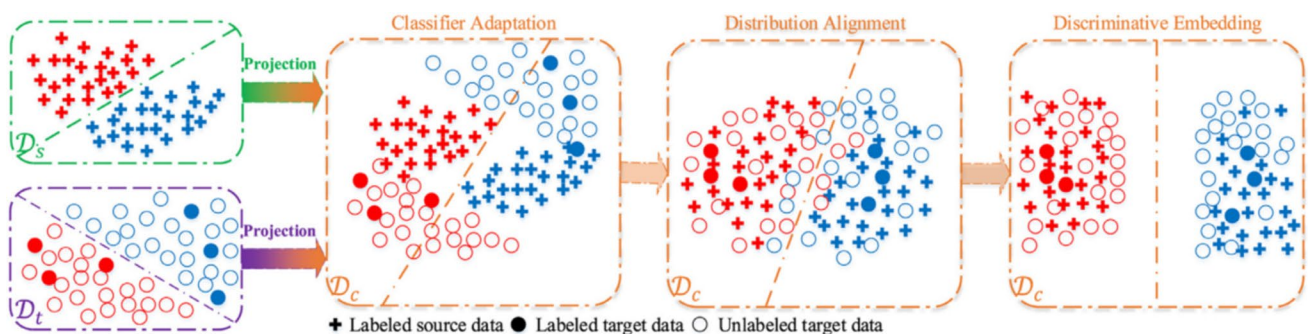
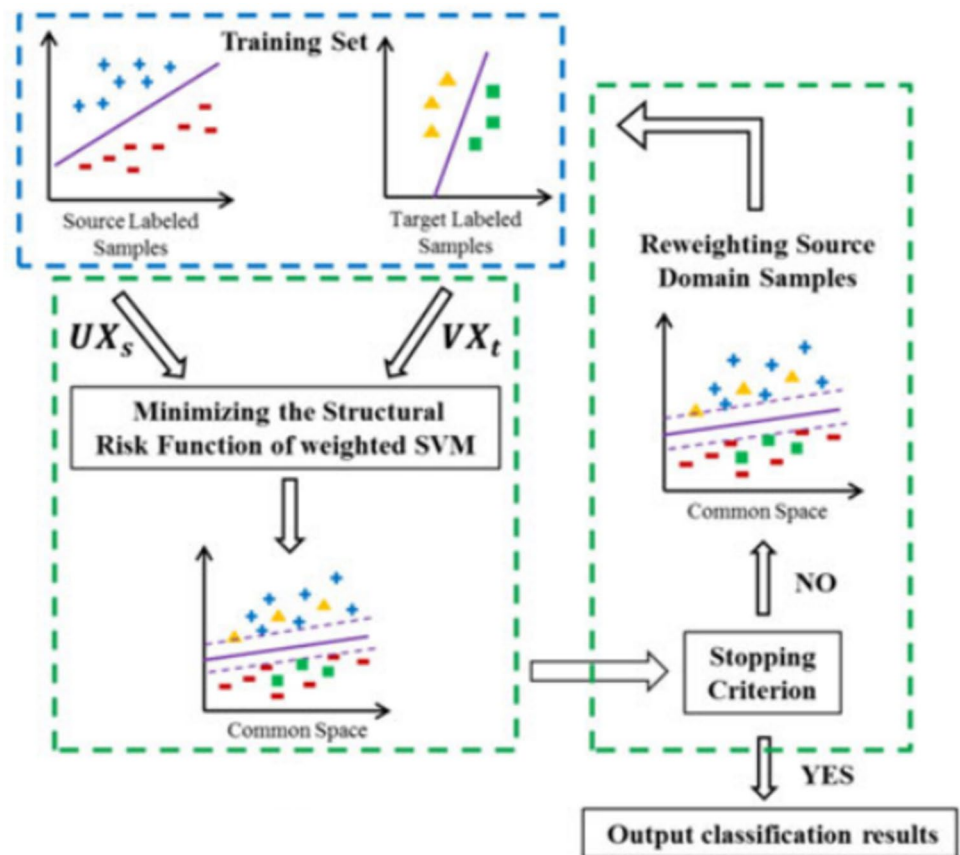
For an unsupervised environment, one of the challenging methods has been presented by authors in the paper (Liu et al. 2020). The authors propose an unsupervised knowledge transfer method that ensures knowledge transfer accuracy and a principal angle-based measurement for determining the distance between two pairs of domains: one pair consists of two homogeneous representations of target and source domains. In contrast, the other pair comprises the source and target domains (Liu et al. 2020).

The research conducted in reference (Zhu et al. 2021) proposes the utilization of a deep sub-domain adaptation network (DSAN) that relies on sub-domain adaptation principles. During unsupervised domain adaptation, DSAN's primary objective is to harmonize the activation distributions within specific sub-domains across various domain-specific layers. By aligning these pertinent sub-domain distributions, DSAN enhances the capability of deep adaptation networks (DANs) to represent features effectively.

One notable progress in sub-domain adaptation lies in its ability to capture detailed information for each category within an end-to-end framework, representing a significant improvement over the previous domain adaptation method. This advancement can be observed in DSAN, as illustrated in Fig. 13. DSAN effectively reduces the disparity between relevant sub-domain distributions of activations in layers L by implementing LMMD minimization.

To overcome the limitations of UOSDA, DSAN introduces a novel measure called Local Maximum Mean Discrepancy (LMMD). LMMD measures how much the kernel mean embeddings of distributions from related sub-domains in the source and target domains differ, using the Hilbert–Schmidt norm to quantify this difference. Notably, LMMD incorporates the influence of various sample weights, allowing it to focus on the most informative and relevant data points.

The authors of the paper (Li et al. 2017) offer IRH transfer learning. This iterative reweighting heterogeneous transfer learning framework learns a shared space for source and target data and reweights source samples using

Fig. 14 The architecture of IRH (Li et al. 2017)**Fig. 15** A block diagram depicting the architecture of DDA (Yao et al. 2020)

a unique iterative reweighting approach. Central to IRH's operation is the iterative reweighting of source data points. Using a weighted support vector machine, the framework first projects both domains into a shared space. It then leverages this shared representation to refine the weights of each source data point, reflecting their relative importance for the target domain, as depicted in Fig. 14.

Heterogeneous domain adaptation tackles the challenge of bridging knowledge gaps between domains with distinct feature spaces, aiming to create accurate models in a target domain by leveraging insights from a source domain.

To solve this kind of problem, Yao et al., as authors in the paper (Yao et al. 2020), present a unified learning method for constructing a domain-invariant subspace called discriminative distribution alignment DDA. The suggested method can adapt discriminative domain directions, align distributions across domains, and improve data separability during adaptation. DDA accomplishes this by minimizing distribution divergence and increasing distances between class centroids to train an adaptive classifier. They design two techniques based on the suggested DDA framework, one

for cross-entropy loss and the other for squared loss. The architecture of the DDA framework is illustrated in Fig. 15.

As a result, the DDA's overall learning structure is as follows (Yao et al. 2020):

$$\begin{aligned} \min_{\phi_s, m_{\phi_t}} & \mathcal{F}[\phi_s(X_s), \phi_t(X_t)] \\ & + \frac{\beta}{2} \mathcal{D}[\phi_s(X_s), \phi_t(X_t)] \\ & - \frac{\tau}{2} \mathcal{I}[\phi_s(X_s), \phi_t(X_t)] \end{aligned} \quad (5)$$

Which $\mathcal{F}(\cdot, \cdot)$, $\mathcal{D}(\cdot, \cdot)$ and $\mathcal{I}(\cdot, \cdot)$ shows the classifier adaptation, distribution alignment, and discriminative embedding constraints, $\phi_s(\cdot) : \mathbb{R}^{d_s} \rightarrow \mathbb{R}^d$ and $\phi_t(\cdot) : \mathbb{R}^{d_t} \rightarrow \mathbb{R}^d$ are two project functions to project source and target into subspace, and β and τ are two positive regularization factors to control the importance impact of $\mathcal{D}(\cdot, \cdot)$ and $\mathcal{I}(\cdot, \cdot)$. The use of deep neural networks, while providing meaningful results, involves a lot of cost and calculations. On the contrary, using the fuzzy system can be helpful in situations where the problem information is insufficient and imprecise.

From this, some methodologies try to use fuzzy systems in transfer learning. In the topic of domain adaptation, fuzzy rules and Takagi–Sugeno models are frequently employed to transfer information between (Behbood et al. 2015, 2014; Zuo et al. 2017, 2018; Yang et al. 2016) since they may demonstrate how knowledge is transferred between domains and derive general fuzzy representations of two domains.

Several transfer learning strategies were proposed by Deng et al. (Deng et al. 2014b, 2013b) and (Yang et al. 2016) based on the Takagi–Sugeno–Kang (TSK) fuzzy model and a Mamdani–Larsen-type fuzzy system to cope with the situations of insufficient data. Additionally, their techniques were used to detect electroencephalogram signals when data was lacking.

In their work, Liu et al. (2018) introduce a novel approach for unsupervised domain adaptation, known as the Fuzzy Heterogeneous Unsupervised Domain Adaptation model (F-HeDUA), which incorporates fuzzy equivalence relations into the context of heterogeneous domains. F-HeDUA represents a significant and challenging advancement in this field.

The core idea behind F-HeDUA is the utilization of fuzzy relations as a metric within a fuzzy space to quantify the similarity between two fuzzy vectors. These fuzzy equivalence relations matrices are then generated for both the source and target domains. Furthermore, the Shared Fuzzy Equivalence Relations model (SFER) is introduced, enabling both domains to share the same number of clustering categories. Their model is based on two key aspects of fuzzy equivalence relations. This approach effectively transfers information and knowledge from the source domain to the target domain via these clustering categories, even when the two domains possess varying quantities of data.

Their research demonstrates how fuzzy algorithms may be used to overcome heterogeneous unsupervised domain adaptation concerns. It proposes a new method (similar to the SFER) for obtaining common representations of two domains using fuzzy technologies. To validate the suggested model and present satisfactory findings, experiments spanning four actual datasets—credit assessment and cancer detection—each made up of two different datasets—were carried out.

Furthermore, classification tasks have received the most attention and significant advancements in both homogeneous and heterogeneous domain adaption, although good solutions to regression issues are limited. Two novel fuzzy rule-based techniques to address regression issues are proposed in research (Zuo et al. 2019b). Fuzzy homogeneous domain adaptation, the first approach, is used for homogeneous areas, whereas fuzzy heterogeneous domain adaptation, the second approach, is used for heterogeneous spaces. A learning algorithm applied to the source domain data extracts its knowledge in the form of fuzzy rules. By generating a latent feature space that decreases the distance between the feature spaces of the two domains, they are moved to the target domain. They illustrate the efficacy of both approaches through experiments on artificial datasets and analyze the significance of several key performance-affecting parameters.

10 Transfer Learning Based On The Number Of Domains

One key aspect of categorizing domain adaptation scenarios is the number of source and target domains involved. This section explores various configurations of transfer learning approaches based on these categories:

- **Single-source and Single-target:** Some papers, such as (Pan et al. 2011), outline domain adaptation approaches that focus on transferring knowledge from one specific source domain to one particular target domain.
- **Multi-source and Single-target:** This scenario involves transferring knowledge from multiple source domains to a single-target domain. References (Jiang et al. 2017), and (Tian et al. 2020) to Li et al. (2014) provide insights into approaches for multi-source adaptation to a single-target domain.
- **Single-source and Multi-Target:** This scenario consists of a single-source domain adapting to multiple target domains, with relevant information found in references (Salimans et al. 2016) and (Liu et al. 2021c) to Liu et al. (2020).

Previous studies have thoroughly investigated the adaptation of single-source and single-target domains. Several works (Zhu et al. 2019c; Saito et al. 2019; Xie et al. 2019b; Nguyen et al. 2020; He and Zhang 2019) focus on decreasing

the feature-level domain difference between the source and target domains through feature alignment techniques. Notably, the major concept in the study (Redko et al. 2019) aligns areas directly relevant to object detection across two domains.

Object Detection Alignment: The major concept in the study (Zhu et al. 2019c) aligns areas directly relevant to object detection across two domains. **SW-Detection Approach:** Proposed by Saito et al. (2019), this strategy aligns local features with strong intensity and global features with broader variations, facilitating domain adaptation across different data sources. This approach effectively bridges the domain gap by focusing on each feature representation level's distinctive characteristics. **Distribution Alignment:** Xie et al. (2019b) tackle the domain adaptation challenge by simultaneously aligning the distributions of local and global features within the same pair of domains. Their method employs a statistical technique known as MMD to measure the difference between feature distributions and adjusts them accordingly, enhancing the overall adaptation process. **Multi-Level Attention Alignment (MEAA):** The MEAA approach (Nguyen et al. 2020) introduces multi-level attention alignment, enhancing the model's capacity to quickly identify structure-invariant items of interest by measuring the uncertainty of each local region through entropy. **Hierarchical Domain Feature Alignment:** The MAF model (Liu et al. 2020) addresses the challenge of domain disparity in feature representation, implementing an automatic hierarchical domain feature alignment paradigm. In addition, Chen et al. (2018b) proposed a domain-invariant Region Proposal Network (RPN), utilizing consistency regularization within the Faster R-CNN model. Xu et al. (2020) introduced a method involving categorical regularization to mitigate domain shifts between two domains. HTCIN (Chen et al. 2020) refines local discriminability by employing a context-aware instance-level alignment module, leveraging the interplay between instance-level features and contextual information. The challenges of data availability in practical scenarios motivate research into data-free unsupervised domain adaptation. Study (Tian et al. 2022) addresses this issue by introducing VDM-DA, an unconventional method for domain adaptation in the absence of source data samples. In this strategy, a virtual domain serves as a bridge between the source and target domains. Virtual domain samples are generated in the feature space using an approximated Gaussian Mixture Model (GMM) based on the pre-trained source model, ensuring that the virtual domain's distribution closely resembles that of the source domain, even without direct access to the source data. In addition to the research discussed in this section, other studies focus on the challenges of single-source learning transfer. While existing methods demonstrate impressive detection performance, their effectiveness may diminish in scenarios involving multi-domain labeled source data characterized by diverse feature distributions. Such scenarios often exhibit several domain gaps

that single-source transfer learning methodologies struggle to overcome.

11 Single Source and Single Target Transfer Learning: Strategies and Applications

The single-source, single-target transfer learning paradigm has proven to be a fertile ground for advancements across diverse learning tasks, including image classification, object segmentation, and complex regression problems. Pioneering works, such as those referenced in Zhao et al. (2018), highlight how this focused setting fosters efficient knowledge transfer, propelling advancements in numerous domains. Domain adaptation is at the heart of this progress, a powerful technique that bridges the gap between source and target data through feature and parameter transformations. Recent research has explored innovative solutions for this crucial step. A notable example is the 'central moment discrepancy' metric, which provides a computationally efficient and accurate means to measure the distance between probability distributions, surpassing earlier methods' cumbersome kernel matrix calculations (Zellinger et al. 2017).

Additionally, the fusion of fuzzy systems with granular computing has shown promise in regression transfer, seamlessly navigating both homogeneous and heterogeneous feature spaces (Li et al. 2021).

Deep learning has emerged as a transformative force in transfer learning, with its intricate neural networks wielding the power to extract domain-agnostic features from diverse sources and targets. This potent ability transcends mere data distributions and label spaces, paving the way for effective knowledge transfer even when faced with stark domain disparities. Recent studies have delved into deep feature transferability nuances, exploring where and what information to share within the complex network architecture. In the realm of multi-representation adaptation, innovative networks have been developed to bridge the gap between source and target representations, drawing upon multiple layers and aspects to achieve superior alignment. Meanwhile, joint geometrical and statistical alignment methods unify shared spaces and subspaces across domains, tackling both geometric shifts and distributional discrepancies simultaneously (Zuo et al. 2019a).

Dynamic distribution adaptation pushes the boundaries further, offering a flexible framework for both classic and deep transfer learning. This approach quantifies the significance of cross-domain marginal and conditional distributions, guiding the knowledge transfer process with exquisite precision. Traditional manifold space transfer extracts Grassmann manifold features, incorporating rich domain details and properties. Geodesic flow kernels minimize computational complexity while traversing

data to the manifold space, and joint distribution weights are computed using geometrical properties governed by Laplacian regularization. Finally, the deep dynamic distribution adaptation network takes centre stage, wielding an end-to-end structure that seamlessly combines the classifier and feature extractor. This potent architecture estimates alignment weights for multi-level distributions, unlocking even finer-grained knowledge transfer through the A-distance metric (Wang et al. 2020b).

12 Transfer Learning With Multi-Source And Single-Target: Strategies and Applications

However, the previous section mentioned experiments concentrated on adaptation to a single-source domain. Traditional single-source adaptation is giving way to the richer picture of multi-source approaches. With target domains often relating to multiple, diverse sources, each brimming with unique transfer potential, the advantages of multi-source adaptation are becoming increasingly clear, propelling its rise in popularity (Zhu et al. 2019a).

To handle numerous sources domain adaption challenges for both classification and regression issues, Zhao et al. (2018) and Wen et al. (2020) quantified the discrepancy by H-divergence. They introduced a deep learning framework powered by an adversarial training approach. The research (Gopalan et al. 2014) offers novel generalization bounds and strategies for unsupervised multi-sourcedomain adaption in both classification and regression tasks. They explain how to understand it as learning feature representations that are invariant to the various domain changes while still being discriminative for the learning goal. Their theoretical analysis naturally leads to an effective learning method employing adversarial neural networks. To do this, they suggest multi-sourcedomain adversarial networks (MDAN), which use a task-adaptive generalization bounds-based approach to domain adaptation. Also, Wen et al. (2020) create a domain discrepancy-based finite-sample generalization bound and, in response, provide a theoretically supported optimization method. Domain Aggregation Network (DARN), their proposed technique, can automatically and dynamically strike a balance between adding more data to improve the effective sample size and removing irrelevant data to prevent detrimental impacts during training.

In real-world applications, Guo et al. (2019) introduced a method that combines experts to perform text classification and speech tagging from diverse sources. They employed the Mahalanobis distance and confidence scores to establish connections between source and target domains. In scenarios with heterogeneous feature spaces, Mahalanobis distance transcends

Euclidean distance, accounting for both mean and covariance differences. This nuanced measurement empowers transfer learning techniques to effectively bridge the gap between domains, facilitating the construction of reliable predictive models even when target data features differ markedly from source data.

In a pioneering study, Ding et al. (2018) devised a novel approach to bridge the knowledge gap between multiple source domains with incomplete information and a target domain. Their strategy hinged on a low-rank matrix that ingeniously filled in missing labels within the source domains by leveraging latent attributes extracted from the target domain. This innovative approach enabled effective domain adaptation even in the face of fragmented source data. Addressing the challenges of disparities between multi-sourcedomains, Redko et al. (2019) focused on multi-sourcedomain adaptation in the presence of target and category shifts, where source labels may not completely align with target domain labels or may do so to varying degrees. As described by Zhu et al. (2019b), a two-stage alignment framework was proposed to address domain mismatch. The initial stage aligns feature distributions across domains, effectively reducing domain-specific differences. The subsequent stage focuses on aligning source-domain classifiers to minimize variations among them, leading to improved performance.

In a recent study (Yin et al. 2022), the researchers introduce the concept of universal multi-source domain adaptation, a broader framework for addressing image classification in situations where multiple source domains possess distinct label spaces and the label set of the target domain remains unknown. The core challenge in universal multi-source domain adaptation lies in establishing a shared label set that is applicable across all source and target domains while maintaining scalability, even as the number of source domains increases. To tackle this challenge, the authors introduce a novel solution, termed the universal multi-source adaptation network (UMAN), designed to handle the domain adaptation problem within a diverse universal multi-source domain adaptation context, without introducing unnecessary complexity to the model.

UMAN leverages a unique pseudo-margin vector and its weighted counterpart to evaluate the reliability of each known class within the common label set. This approach assists in aligning the data distributions between various source and target domains through adversarial training. Additionally, UMAN offers a theoretical underpinning for its efficacy. Algorithm 1 illustrates the UMAN algorithm in detail. This innovative approach addresses the complexities associated with universal multi-sourcedomain adaptation, offering a practical and efficient solution for image classification across diverse domains.

Input: T: maximum iteration

w_0 : the condition for discriminating between samples of unknown and known classes;

$D_{s_i} = \{(x_j^{s_i}, y_j^{s_i})\}_{j=1}^{n_{s_i}}$ $i = 1, 2, \dots, M$: source training datasets; $D_t = \{(x_k)\}_{k=1}^{n_t}$ target training dataset

F : pre-trained feature extractor parameterized by θ_f ;

G : randomly initialized classifier parameterized by θ_g ;

D : randomly initialized domain classifier parameterized by θ_d ;

Output: θ_f , θ_g and θ_d .

Training:

set $t = 0, \epsilon = 0.1$ and initialize $V_{TMR} = V_{TMR}^0$;

while $t < T$ **do**

for each batch $(D_{s_1}^b, D_{s_2}^b, \dots, D_{s_M}^b, D_t^b)$ **in** $(D_{s_1}, D_{s_2}, \dots, D_{s_M}, D_t)$ **do**

 obtain extracted features on each domain: $z_j^{s_i} = F(x_j^{s_i})$ and $z_k = F(x_k)$

 obtain classification results using SoftMax $G(x_j^{s_i}) = G(z_k)$

 calculate the pseudo-margin, source error, weight $x_j^{s_i}$ and weight x_k

 calculate joint cross entropy loss E_G and E_D

update θ_f , θ_g and θ_d

 let $t=t+1$

Testing:

Test data x is forwarded to obtain $G^c(F(x))$ over $c \in C_c$;

If $G \circ F(x) \geq w_0$ **then**

 label x with $y = \arg \max_c G^c(F(x))$;

else

 reject x as an unknown class

end

In the context of multi-source domain adaptation, (Liu et al. 2021c) introduced a groundbreaking approach that leverages deep neural networks to achieve significant performance breakthroughs. This innovative method employs distribution fusion and relationship extraction to learn the weights of multiple sources, ultimately enhancing predictor performance in the target domain. By determining how closely the sources are associated with the target during training, as discussed in Li et al. (2018), the approach ensures that sources most comparable to the target task contribute more effectively to the learning process. Transfer learning can be conceptualized as a bridge for object recognition algorithms, facilitating their traversal across the diverse landscapes of different domains. This metaphorical bridge is being actively constructed and fortified, driven by the growing interest in this powerful technique. Historically, most existing algorithms have concentrated on unsupervised domain adaptive object detection from a single-source domain. However, practical applications often involve labeled data from various domains with diverse feature distributions, significantly increasing complexity and grounding challenges in realistic settings. Traditional strategies typically struggle in these scenarios due to the presence of multiple domain gaps.

M Source domains

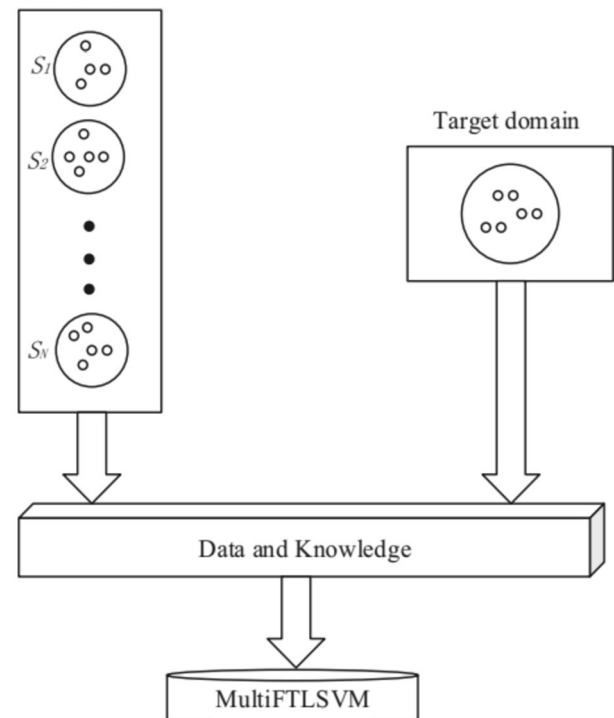


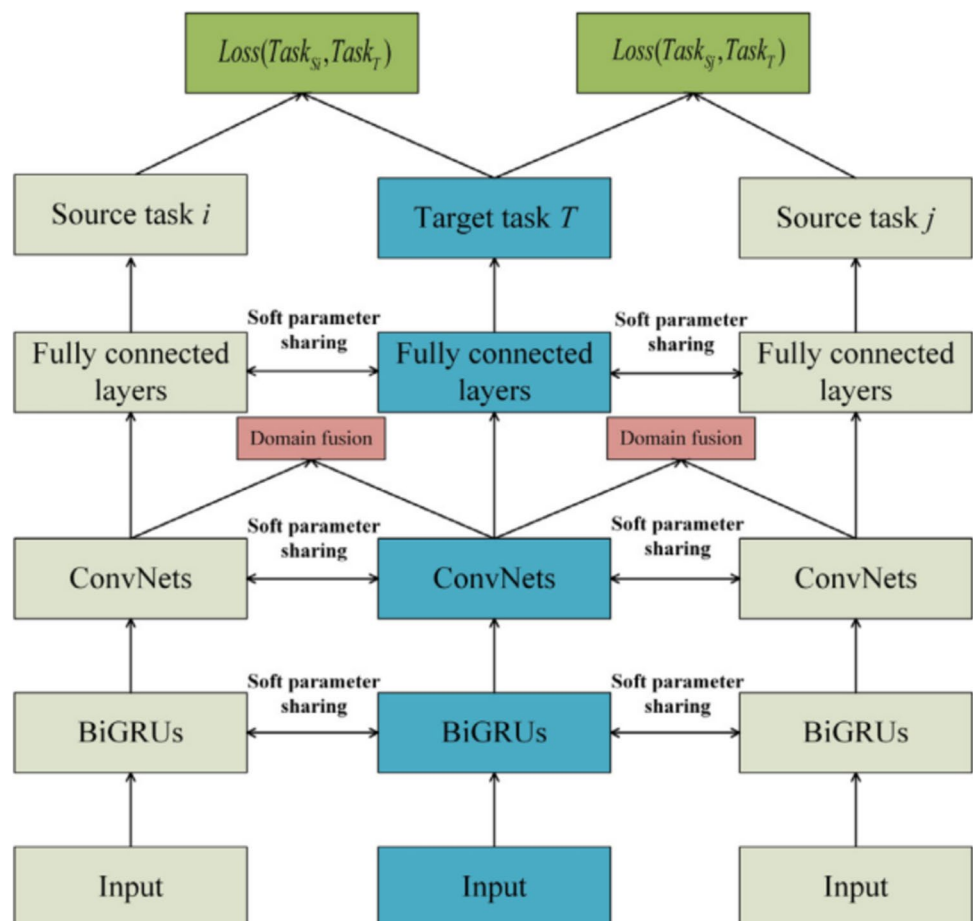
Fig. 16 The architecture of MultiFTLSVM(Gao et al. 2021)

To address these challenges, Zhang et al. (2022b) proposed a multi-source domain knowledge transfer approach called MKT. This methodology comprises four key steps designed to enhance domain adaptation: **Alignment of Local Domain Features:** This initial step focuses on aligning domain-specific features using a domain discriminator after training a shared local feature extraction network. **Alignment of Distributions within Specified Domains:** A multi-branch network is constructed, with each branch dedicated to extracting features from a specific source domain. Corresponding domain discriminators for each branch facilitate the alignment of features between the source and target domains. **Data Combination:** High-level features from various data sources are mapped into a shared space, allowing for an evaluation of each sample's potential for cross-domain application. This transferability assessment subsequently informs the training of a unified detection network for all branches. **Attention Mechanism:** An attention mechanism is incorporated to enhance model identification performance in the target domain. Both image-level and instance-level attention techniques are employed to focus on relevant aspects of the data, ensuring that the model concentrates on critical features during the learning process.

In summary, the multi-source domain knowledge transfer approach developed by Zhang et al. (2022b) provides a comprehensive solution to the challenges of domain adaptation in object recognition. This method aims to enhance model performance in scenarios characterized by diverse source domains and feature distributions.

Building on this foundation, Gao et al. (2021) proposed a novel transfer learning algorithm called Multi FTLSTM, based on support vector machines. This algorithm leverages knowledge from multiple sources for transfer and is designed to be faster than existing alternatives. The technical details of the proposed method's architecture are illustrated in Fig. 16. The central idea of this approach is to improve classification effectiveness by utilizing multiple sources rather than relying on a single-source. This is achieved by simultaneously incorporating all data from the source domains, which expedites the training process and enhances overall efficiency. The initial step involves calculating similarity weights between source and target samples. These weights are then used to mitigate discrepancies in marginal probability distributions, effectively preventing negative transfer, where the knowledge from the source domains adversely affects the target domain's performance. Overall, this innovative

Fig. 17 The architecture of MDAJL (Zhao et al. 2020)



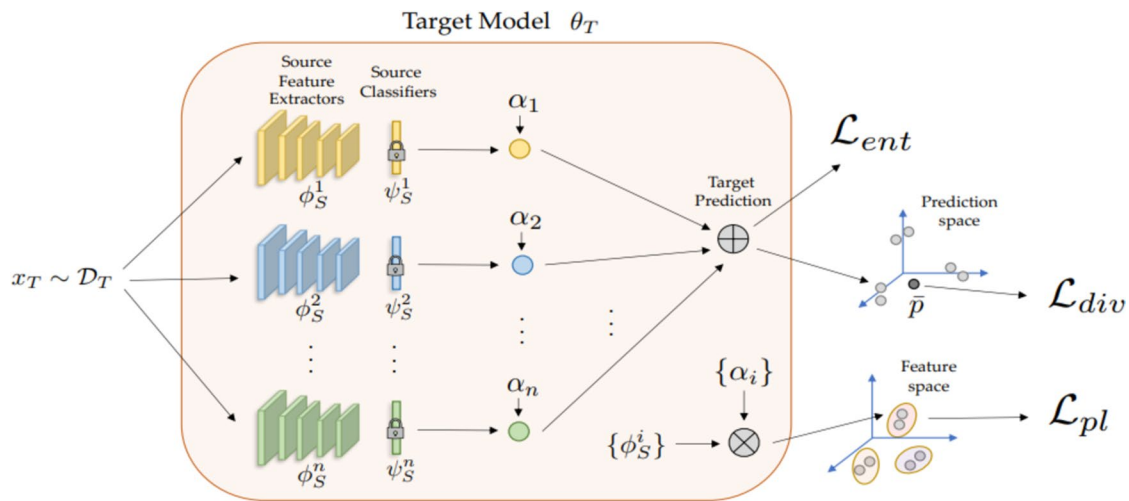


Fig. 18 The architecture of DECISION (Ahmed et al. 2021)

algorithm contributes significantly to the advancement of multi-source domain adaptation techniques, offering a faster and more effective means of enhancing model performance in diverse scenarios.

The approximation support vector machine is then utilized to gather sample datasets and related weights from each source domain that are relatively significant to model training, improving the algorithm's training efficiency. Finally, knowledge from the target domain and combinatorically weighted multi-sourcedomains are pooled and incorporated into the support vector machine's structural risk reduction architecture. After that, an objective function is created, and a theoretical demonstration is carried out. MultiFTLSVM's efficacy has been demonstrated on various real datasets and offers several benefits over the benchmark technique.

In cross-domain classification, information in source domain tasks is used to enhance the classification of the task. It can significantly boost the use of labeled data in source domains while minimizing data integration efforts in the new domain. Currently, available techniques rely on information from a single-source domain and hard parameter-sharing methods, which are likely to overlook discrepancies between domain-specific features. A groundbreaking framework, MDAJL, has been developed by Zhao et al. (2020) to advance sentiment classification capabilities in multi-source, cross-domain settings with multi-sourcedomain adaptation and collaborative learning. Figure 17 visualises their proposed technique, revealing how it extracts important features from data while strategically sharing parameters between different tasks. This approach employs bi-directional gated recurrent units and convolutional neural networks. Furthermore, deep domain fusion reduces distance requirements. Multiple concurrent task learning is used in

multi-source domain adaptation, and the gradients are back-propagated simultaneously.

However, most current unsupervised domain adaptation approaches include the strong assumption that the source data would be available during training, which may not be practicable owing to privacy, security, and storage considerations. A recent breakthrough introduced an algorithm capable of transferring knowledge from single or multiple source models to unlabeled target domains, without relying on access to the source data.

Ahmed et al. (2021) created a novel unsupervised domain adaptation algorithm that can learn from and merge numerous source models in the best possible way without the need for source data. They present theoretical foundations for their approach and demonstrate its efficacy in several domain adaption benchmarks.

The overall framework of their technique is as Fig. 18. They propose to freeze all of the sources' final classification layers and jointly optimize for the source feature encoders and their related weights to achieve the target predictor by merging them. In Fig. 6, θ_T is the target model, which maps input space, $\mathcal{X}$ to $\mathbb{R}^K$, K is the number of K categories they split each source model into two sub-modules: the feature extractor as $\Phi_s^i \rightarrow \mathbb{R}^{d_i}$ and the classifier $\Psi_s^i \rightarrow \mathbb{R}^K$. Here, d_i shows the feature dimension of the i -th model while K refers to the number of categories also α_i is the weight of each source model. In summary, the final aim of the authors is to minimize the $\mathcal{L}_{total}$ as:

$$\mathcal{L}_{total} = \mathcal{L}_{ent} - \mathcal{L}_{div} + \lambda \mathcal{L}_{pl} \quad (6)$$

The aim is to minimize the $\mathcal{L}_{ent}$ that is the conditional entropy term, maximizing the $\mathcal{L}_{div}$ as the entropy of the

empirical distribution of labels and reduce the $\mathcal{L}_{pl}$ as the cross-entropy loss of pseudo-labels.

Despite their significant contribution to developing deep neural networks, the lack of interpretability of transfer learning algorithms is a highly contentious issue (Tan et al. 2015). The popularity of fuzzy systems has grown due to their simplicity and capacity for an explanation. Furthermore, fuzzy models are effective in situations of uncertainty due to their ability to handle imprecision, vagueness, and ambiguity. As a result, with their ability to handle uncertainty and incorporate expert knowledge, fuzzy systems offer a valuable tool for exploring transfer learning solutions. Consequently, investigations have been conducted into the problems with multiple-source domain adaptation. As described by Yao and Doretto (2010), TaskTraAdaBoost and Multi-source-TrAdaBoost enhance the boosting paradigm by introducing mechanisms for multi-source knowledge transfer, enabling models to leverage information from multiple datasets with potentially different characteristics. By incorporating additional information, this approach mitigates the impact of negative transfer, leading to improved performance.

By enabling various views from multiple sources to complement one another through a co-training style framework, Tan et al. (2014) introduced a unique method to jointly harness information from many viewpoints and sources and lessen the discrepancies in distribution across various domains. When transferring knowledge from several source domains to the target domain, Tan et al. (2014) found that doing so produces a more potent feature representation of the data than the source data. In this transfer learning approach, several classifiers from the source domain data (e.g., text classification) are jointly trained with auto-encoders to learn a shared, hidden representation and domain-specific classifiers simultaneously. The auto-encoders, specifically variational auto-encoders, are trained with a reconstruction loss to learn a compact and informative representation of the source data. The classifiers then use this representation, trained with a cross-entropy loss, to perform classification tasks on the target domain (e.g., image classification). These methods were created for classification challenges, not domain adaptation difficulties. It has not yet been possible to use knowledge from different sources to create fuzzy systems that are better at handling uncertainty.

One problem with fuzzy transfer learning hasn't been solved, even in general transfer learning: how to aggregate and apply the information when several source domains are accessible. Building on existing research, (Lu et al. 2020b) proposes novel techniques for fusing fuzzy rules from various domains in regression tasks. The study explores two distinct scenarios: homogeneous and heterogeneous spaces. In homogeneous feature settings, knowledge from different sources is combined as fuzzy rules. Conversely,

heterogeneous feature settings integrate knowledge through both data and fuzzy rules.

Transfer learning is a powerful technique that utilizes knowledge acquired from one domain to enhance learning in another domain. It can be categorized into inductive transfer learning, which focuses on different tasks between domains, and Transductive transfer learning, which deals with the same tasks but different domains. Inductive transfer learning methods include instance-based and feature-based approaches, such as TrAdaBoost and feature extraction. Transductive transfer learning often employs feature-based techniques to predict label distribution in the target domain using source domain data. Transfer learning can occur in a homogenous space, where the feature dimensions are the same, or a heterogeneous space, where they differ. Homogeneous transfer learning strategies involve modifying marginal and conditional distribution differences through subspace-based or deep feature learning methods. Semi-supervised and unsupervised techniques, such as TCA and scatter component analysis, are used in homogenous domain adaptation. Heterogeneous transfer learning aims to bridge the gap between feature spaces and can be performed in supervised, semi-supervised, or unsupervised settings. Various methods address heterogeneous transfer learning challenges, including HeMap and unsupervised knowledge transfer. Fuzzy systems can be applied in both homogenous and heterogeneous scenarios to handle imprecise or insufficient information. Transfer learning based on the number of sources includes single-source and single-target, multi-source and single-target, and single-source and multi-target adaptation. Techniques focus on reducing domain differences, discovering shared latent spaces, quantifying domain discrepancy, and balancing data contributions. Universal multi-source domain adaptation and multi-output regression transfer learning handle challenges like different label sets and joint prediction of

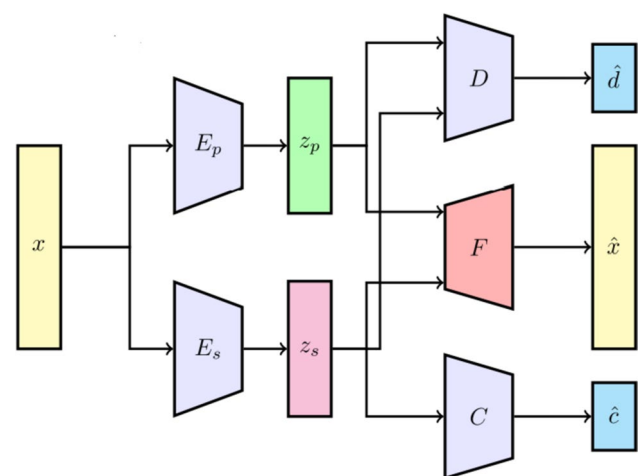


Fig. 19 The architecture of MTDA-ITA (Gholami et al. 2020)

Table 2 Brief introduction of datasets

Dataset	Domains	Instances	Features	Classes	Cross Domain
PIE	PIE-A	3332	1024	68	20
	PIE-B	1629			
	PIE-C	1632			
	PIE-D	3329			
	PIE-E	1632			
Office-31	DSLR	498	800 (SURF)	31	6
	Amazon	2817	4096 (DeCAF ₆)		
	Webcam	795			
Office-Caltech	DSLR	1410	800 (SURF)	10	12
	Amazon	1123	4096 (DeCAF ₆)		
	Webcam				
	Caltech				
Office Home	Art	2427	2048 (ResNet50)	65	12
	Clipart	4365			
	Product	4439			
	Real World	4357			
Digits	MNIST	2000	2048 (CNN Features)	10	2
	USPS	1800	256		
Amazon Review	Books	340,000	5000	2	12
	DVD				
	Electronics				
Reuters-21578	Kitchen				
	Orgs	21,587		5	20
	People				
	Places				
	Topics				
Exchange					

multiple outputs. Object recognition also explores transfer learning for multi-sourcedomain adaptation, while some algorithms enable unsupervised domain adaptation without source data access. Transfer learning techniques offer versatile approaches for leveraging knowledge from related domains to improve learning performance.

13 Transfer Learning With Single-Source And Multi-Target: Strategies And Applications

Models of unsupervised domain adaptation concentrate on paired adaptation scenarios with a single-source domain with labeled data in front of a single-target domain with no label. However, one must adapt to various but comparable target domains in many real-world situations. Pair-wise adaptation procedures may be poor because they do not use common knowledge across several domains. In the work (Gholami et al. 2020), Gholami et al. propose a novel domain adaptation technique to bridge the gap between multiple unlabeled target domains and a single labeled source domain. Their method strives to establish a universal latent space encompassing all domains while accommodating their distinct characteristics.

The authors utilize a cohesive information-theoretic framework to differentiate between shared and

domain-specific information, thereby improving the association between latent representations and observed data. Supported by an effective optimization technique, their model facilitates concurrent adaptation from a single-source to multiple target domains. This accomplishment is realized by augmenting the mutual information between private (domain-specific) features and domain labels while concurrently diminishing the mutual information between shared (domain-invariant) features and domain labels. Figure 19 shows the MTDA-ITA method the authors proposed in the paper (Gholami et al. 2020). Based on this framework, the encoder $E_s(x)$ encodes the feature representations (z_s) that are common across domains for a given input sample x . Using the shared private encoder, $E_p(x)$ extracts domain-specific private features (z_p). Using both private and communal features, the shared decoder $F(z_p, z_s)$ learns to rebuild the input sample.

The domain classifier $D(z_s/z_p)$ learns to successfully predict the domain labels of real samples based on both common and private features, whereas the classifier $C(z_s)$ learns to correct transfer learning and predict class labels on shared features.

Current transfer learning techniques cannot solve multi-output regression transfer learning problems. A straightforward and uncomplicated strategy involves employing regression transfer learning models to forecast multiple outputs

Fig. 20 Example images of the PIE dataset (Pereira and Torres 2018)



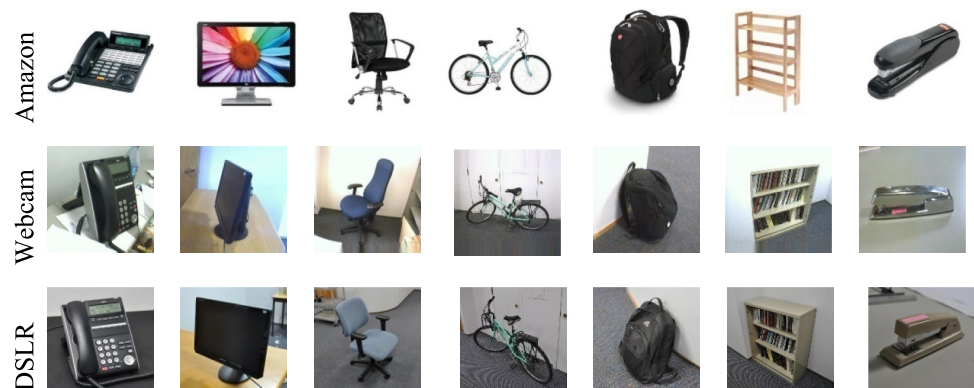
collectively. Prior research (Che et al. 2021) has tackled the challenges of multi-output regression in both homogeneous and heterogeneous scenarios through an innovative fuzzy transfer learning methodology. This method involves deducing fuzzy rules that mirror the shared characteristics of various outputs, capturing their distinctiveness through considering output-input relationships, and addressing inter-output correlations.

Initially, fuzzy rules are compiled within a source domain with ample training data tailored for a homogeneous scenario. These fuzzy rules are subsequently employed in fresh yet related regression tasks within the target domain. In this context, the target domain is characterized by limited data but numerous responses, employing diverse transformation methodologies. Building upon this groundwork, we extend the methodology to handle a more intricate heterogeneous scenario. The method involves extracting a common, underlying representation of the data that transcends domain-specific differences, making variables more compatible.

14 Dataset And Result

Transfer learning addresses critical challenges in machine learning, such as the discrepancies in the distribution of labeled samples and the scarcity of accessible labeled data. These issues often hinder the performance of traditional machine-learning models. As a result, transfer learning techniques have gained traction, demonstrating success in numerous real-world applications. To comprehensively evaluate the efficiency and effectiveness of the proposed method, the authors implemented it across various datasets. This implementation underscores the method's performance in diverse contexts and challenges. Several of these datasets are publicly available and pertain to two significant areas of study: text classification and object recognition. This section begins with an overview of the datasets frequently utilized in relevant literature. Subsequently, we will introduce the validation criteria employed for assessment, followed by a comparative presentation of empirical findings.

Fig. 21 Exemplary Images from the Office-31 Dataset (Saenko et al. 2010)



15 Dataset

Datasets selected for transfer learning tasks can vary based on the specific classification or regression objectives. For broader classification applications, datasets can be categorized into text and image categories. The most commonly used datasets in transfer learning experiments include PIE, Amazon Reviews, and Office-31. Table 2 provides a summary of the preprocessed datasets used in our study.

PIE is the Pose, Illumination, and Expression (PIE) database (Pereira and Torres 2018) at Carnegie Mellon University (CMU). It is a standard and well-known database for face recognition. It consists of 41,368 grayscale facial images taken from 68 people as a subject in 32×32 , each of whom is represented in 13 different postures in front of synchronized cameras and four emotions. The five subsets are left, up, down, front, and right pose. Each subset is chosen as an experimental domain, each corresponding to a different pose: PIE-A consists of images in the left pose, PIE-B consists of images in the up pose, PIE-C comprised of images in the downward pose, PIE-D consists of images in front pose and PIE-E comprised of images in the right posture. A random sample of the images is shown in Fig. 20, which is in five domains: PIE-A, PIE-B, PIE-C, PIE-D, and PIE-E, with eight labels.

Office-31 is a standard dataset for object recognition with 31 labeled categories. Images of objects have been collected from three different sources. These three domains have 2817, 498, and 795 instances, respectively. The graphics on Amazon are images from Amazon.com's online store. Webcams' low-resolution photographs are displayed in the Webcam. And the high-resolution photos captured by DSLR cameras are those found in DSLR. Each object was photographed three times from different perspectives. Figure 21 shows a

random sample of the images. So, based on the dataset, six tasks can be considered in single-source transfer learning scenarios (Saenko et al. 2010).

Amazon $\rightarrow$ DSLR ($A \rightarrow D$), Amazon $\rightarrow$ Webcam ($A \rightarrow W$), DSLR $\rightarrow$ Webcam ($D \rightarrow W$), ...

On the other hand, three can be defined in multi-source-transfer learning:

Amazon and DSLR $\rightarrow$ Webcam (A and $D \rightarrow W$),

Amazon and Webcam $\rightarrow$ DSLR (A and $W \rightarrow D$),

DSLR and Webcam $\rightarrow$ Amazon (D and $W \rightarrow A$),

Office-Caltech dataset combines the Office 10 and Caltech10 datasets, providing a unique resource for transfer learning tasks. Compared to the Office 31 dataset, the Office-Caltech dataset stands out by featuring samples labeled in 10 categories. It encompasses three primary domains: Amazon, DSLR, and Webcam. Notably, it introduces an additional domain called Caltech, characterized by images captured in a more authentic environment, enriching the dataset with diverse visual representations. In total, the dataset comprises 2533 images.

In a single-source transfer learning scenario, the Office-Caltech dataset facilitates 12 distinct tasks, allowing practitioners to leverage its labeled samples across domains for various learning objectives. On the other hand, six unique tasks can be defined in multi-source transfer learning environments, where information from multiple domains is integrated. This dataset's versatility makes it a valuable resource for exploring and evaluating transfer learning algorithms in both single and multi-sourcesettings, offering a rich set of challenges and opportunities for advancing the field.

The office-Home dataset is a substantial collection comprising 15,588 images designed for object recognition tasks. Encompassing a diverse set of 65 objects, the dataset spans four distinct domains, each providing unique challenges and perspectives. These domains consist of:

Fig. 22 Example images of the Office-Home Dataset

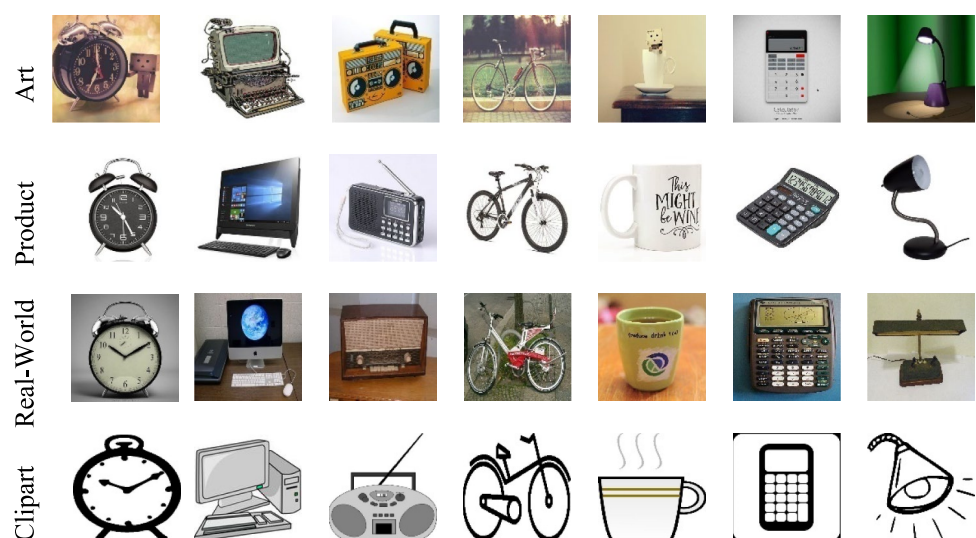


Fig. 23 Example images of the MNIST/USPS dataset(Pereira and Torres 2018)



- Real-World images: Objects photographed with high-quality cameras in real-world scenarios.
- Clipart images: Digitally created images of objects, often featuring simplified and stylized representations.
- Product images: Depictions of objects without a background, emphasizing the objects themselves.
- Artistic pictures: Varied representations, including ornamentation, paintings, sketches, and other forms of artistic renderings.

Each class within the dataset, corresponding to individual objects, contains an average of 70 to 99 images. This diversity ensures a robust and comprehensive representation of the objects in different contexts and artistic styles. Figure 22 provides a visual glimpse into the dataset, showcasing a random sample of images across the four domains: Art, Product, Real-World, and Clipart. The inclusion of such varied domains makes the Office-Home dataset a valuable resource for training and evaluating object recognition models under diverse and challenging conditions.

MNIST/USPS: They are two well-known standard digital datasets with ten handwritten numbers ranging from 0 to 9, meaning they belong to 10 classes. MNIST (Zhong et al. 2021) includes handwritten numbers collected from US high school students and US tax officials and has 70,000 test and train images of size 28×28 . In contrast, USPS (Pereira and Torres 2018) includes handwritten numbers scanned by the US Postal Service and has 9,292 images, divided into two subsets: 7,291 training images and 2,007 test images of size 16×16 . The distributions of MNIST and USPS are considerably different, so they are good databases for domain adaptation and transfer learning. A sample of the images is shown in Fig. 23, which are in two domains, USPS and MNIST, with nine labels.

Amazon Review is one of the datasets used to evaluate text and emotions in transfer learning and machine learning algorithms. It contains 340,000 comments about 25 different products from the Amazon website from 1996. These comments include various products such as electronics, DVDs, books, and kitchen equipment (Wang et al. 2020a).

One of the advantages of the dataset is that it consists of a star rating between zero and five stars and can be

converted to binary numbers if needed. Comments with more than three stars are labeled positive, and comments with less than four stars are labeled negative. Each product review is a separate domain while implementing transfer learning algorithms. It indicates that it can be a source or target domain. For example, the comments on the book can be the source domain, and comments on DVDs can be the target domain. In addition, some studies using deep neural networks have extracted 5,000 repetitive words as features and formed the feature vector to increase efficiency.

Reuters-21578 documents were collected, indexed in 1987, and then categorized by Reuters staff. It was published in 1990 for research purposes and includes 21,587 samples in five categories: people, organs, exchange, places, and topics, which can be used as five different domains. The dataset is categorized as a hierarchical structure, and the task is the prediction of the root (parent) sub-category in the structure. Also, some studies have provided a preprocessed version to increase workload and reduce computational volume (Long et al. May 2014).

16 Experiment Setting

Root Mean Square Error (RMSE) is a commonly utilized metric within regression scenarios, serving as an assessment parameter for measuring the deviation between a model's predicted values and the actual observations within a dataset. The RMSE, calculated as the square root of the mean squared error, quantifies the average magnitude of discrepancies between predicted and observed values, revealing the model's overall predictive accuracy. This process involves squaring the individual errors to ensure that both positive and negative deviations contribute proportionally to the overall error. The square root is then taken to bring the error metric back to the original scale of the data. Mathematically, the RMSE is characterized as follows:

$$RMSE = \frac{1}{n} \sqrt{\sum_{i=1}^n (x_{real} - x_{predicted})^2} \quad (7)$$

Here, n represents the number of data points in the dataset, x_{real} is the observed or actual value, and $x_{predicted}$ is the predicted value for the i -th data point. The RMSE provides a comprehensive measure of the overall model performance in predicting continuous outcomes, making it suitable for regression-type tasks. It is particularly useful in scenarios where an accurate estimation of the magnitude of prediction errors is crucial, such as calculating numerical values like house prices, temperature, or other continuous variables.

In a classification task, the accuracy is calculated by counting the number of instances in the target domain D_T where the model's predictions $f(x_i)$ match the true labels y_i , and then dividing this count by the total number of instances in the target domain. This provides a measure of how well the model performs in terms of correct predictions in the specific target domain of interest.

$$Accuracy = \frac{|\{x|x_i \in D_T \wedge f(x_i) = y_i\}|}{D_T} \quad (9)$$

17 Experiment Result

This section aims to compare the performance of some of the methods introduced earlier on the datasets presented in the previous section. The analysis evaluates the results obtained from these methods, with all algorithms utilizing default or recommended parameter settings as specified in the original papers. Table 3, Table 4 and Table 5 are provided for reference but are not used as a criterion for comparing the algorithms, given the consistent use of default parameters.

The preprocessing stage plays a crucial role in implementing transfer learning algorithms. Some researchers utilize datasets based on the DeCAF and SURF algorithms (Long et al. 2014; Tian et al. 2020), where the feature learning step has been performed. Other studies leverage collections with features extracted using ResNet50 (Zhu et al. 2021; Zhu et al. 2019a). Additionally, certain researchers directly incorporate the feature extraction step into their designed algorithms. The proposed algorithms address various limitations in the proposed method and implementation, enhancing the robustness and efficacy of transfer learning algorithms.

Table 3 encapsulates accuracy performance results for several domain adaptation methods applied to the Office-Home dataset across four distinct domains: Art (A), Clipart (C), Product (P), and Real World (R). Each row corresponds to a specific domain adaptation method, including CLA(Yang et al. 2022), CMMS(Tian et al. 2020), VDM-DA(Tian et al. 2022), GMTL(Zhao et al. 2021), DAOD(Fang et al. 2021) (ResNet50) (Zhu et al. 2019a), SSD(Zhang and

Table 3 Performance Results for Accuracy in the Office-Home Dataset Across Three Domains

	A → C	A → P	A → R	C → A	C → P	C → R	P → A	P → C	P → R	R → A	R → C	R → P	Average
CLA(Yang et al. 2022)	66.7	85.6	90.9	75.6	76.8	86.8	78.7	67.3	88.7	81.6	66.8	87.8	79.4
CMMS(Tian et al. 2020)	56.2	80.8	82.8	65.9	78.7	82.2	67.7	54.5	82.9	69.5	57.1	85.2	72
VDM-DA(Tian et al. 2022)	59.3	75.3	78.3	67.6	76	75.9	68.8	57.7	79.6	74	61.1	83.6	71.4
GMTL(Zhao et al. 2021)	55.4	88.4	95	70.1	84.4	90.9	72.7	54.4	94.7	80.5	56.2	88.7	77.6
DAOD(Fang et al. 2021)	55.3	72.6	78.2	59.2	70.8	77.8	66.7	60	84.1	71.3	58.4	81.8	69.6
(ResNet50)(Zhu et al. 2019a)	53.8	68.6	75	57.3	68.5	68.3	58.5	54.6	77.5	70.4	60	82.2	66.2
SSD(Zhang and Davison 2021)	54.1	80.1	80.9	68.2	81.1	80.1	69.9	55.8	82.1	72.9	57.2	84.3	72.3
DSAN(ResNet50)(Zhu et al. 2021)	54.4	70.8	75.4	60.4	67.8	68	62.6	55.9	78.5	73.8	60.6	83.1	67.7

Table 4 Performance Results for Accuracy in the Office-31 Dataset Across Three Domains

	A → D	A → W	D → W	D → A	W → A	W → D	Average
VDM-DA(Tian et al. 2022)	93.2	94.1	98	75.8	77.1	100	89.7
DAOD(Fang et al. 2021)	89.8	84.2	96	71.8	72.9	97.5	85.7
(ResNet50)(Zhu et al. 2019a)	86.4	91.4	96.9	68.3	70.9	99.8	85.6
CatDA(Wang et al. 2020a)	90.8	87.9	91.1	72.2	74.1	91.4	84.5
GFK(Li et al. 2020a)	74.5	72.8	95	63.4	61	98.2	77.5
CLA(Yang et al. 2022)	100	100	100	94.5	96.6	100	98.5
CMMS(Tian et al. 2020)	72.9	74.7	60.7	97.6	60.3	99.6	77.6
DSAN(ResNet50)(Zhu et al. 2021)	90.2	93.6	98.3	73.5	74.8	100	88.4

Davison 2021), and DSAN(ResNet50) (Zhu et al. 2021). The accuracy scores, expressed as percentages, represent the success of each method in adapting from one domain to another. For instance, the $A \rightarrow C$ indicates the accuracy achieved when adapting from the Art domain to Clip art, and the Average provides a comprehensive view of the average accuracy across all domain adaptation tasks for each method.

An in-depth analysis of the table reveals notable trends in the performance of domain adaptation methods. Certain methods, such as CLA (Yang et al. 2022) and GMTL(Zhao et al. 2021), demonstrate higher overall accuracy averages, suggesting their efficacy in handling diverse domain shifts. Conversely, others, like DSAN(ResNet50) (Zhu et al. 2021), exhibit more varied performance across adaptation tasks. These findings contribute valuable insights into the strengths and weaknesses of different domain adaptation techniques, aiding researchers and practitioners in selecting suitable approaches for specific domain transfer scenarios in image classification tasks.

Table 4 displays the accuracy performance outcomes for the Office-31 dataset across three distinct domains: Amazon (A), DSLR (D), and Webcam (W). Each row corresponds to a particular transfer learning method, while each column signifies a specific domain adaptation task. The table entries indicate accuracy percentages achieved by each algorithm for the corresponding tasks. The columns $A \rightarrow D$, $A \rightarrow W$, $D \rightarrow W$, $D \rightarrow A$, $W \rightarrow A$, and $W \rightarrow D$ specify particular domain adaptation assignments within the Office-31 dataset. For instance, $A \rightarrow D$ signifies the transfer of knowledge from Amazon to DSLR. The Average column provides the mean accuracy across all tasks for each transfer learning method, serving as a comprehensive performance metric. In the VDM-DA(Tian et al. 2022), the value 93.2 in the $A \rightarrow D$ column indicates that the algorithm achieved 93.2% accuracy in transferring knowledge from Amazon to DSLR. In the DAOD(Fang et al. 2021), the value 89.8 in the ' $A \rightarrow D$ ' column indicates an accuracy of 89.8% for transferring knowledge from Amazon to DSLR using the DAOD method. This table serves as a comparative analysis of the effectiveness of various transfer learning methods specifically tailored to the Office-31 dataset and its domain adaptation tasks.

In Table 5, the GMTL (Zhao et al. 2021) row, the value 57.5 in the $C \rightarrow A$ column indicates that the algorithm achieved 57.5% accuracy in transferring knowledge from Caltech Fig. 23 to Amazon. Also, in the SSD(ResNet50) (Zhang and Davison 2021) row, the value 96.2 in the $C \rightarrow A$ column indicates a high accuracy of 96.2% for transferring knowledge from Caltech to Amazon using the ResNet50 model. The "Average" column assesses each algorithm's performance across all domain adaptation tasks. This table allows researchers and practitioners to compare the effectiveness of different transfer learning

Table 5 Performance Results for Accuracy in the Office-Caltech Dataset Across Four Domains

	C → A	C → D	C → W	A → C	A → D	A → W	D → A	D → C	D → W	W → A	W → C	W → D	Average
GMTL(Zhao et al. 2021)	57.5	48.7	44.6	46.7	43.2	43.9	38.6	34.2	86.8	37	33.9	89.5	50.4
SSD(ResNet50)(Zhang and Davison 2021)	96.2	96.2	96.2	94.9	98.3	98.3	96.2	96.2	99.3	98.1	98.1	98.7	97.2
CatDA(Wang et al. 2020a)	44	50.6	49.5	39	46	47.4	39.3	31.1	72.2	41.5	33	68.3	46.9
GFK (Li et al. 2020a)	46.1	55	57	39.6	50.9	56.9	46.2	33.9	80.2	46.2	32.8	75	51.6
DDA(Yao et al. 2020)	47.4	59.2	63.5	31.6	59.5	63.1	-	-	-	46.35	31.83	59.69	51.36
CMMS(SURF)(Tian et al. 2020)	61	51	61.7	39.4	53.5	56.3	46.7	31.9	86.1	40.1	35.8	89.2	54.4
CMMS(DeCAF)(Tian et al. 2020)	94.1	95.5	91.9	88.8	95.5	92.2	93.4	89.3	99.3	93.8	89	100	93.6
TJM(ResNet50)(Long et al. 2014)	94.7	86.8	86.6	83.6	76.4	82.7	92.5	87.4	98.3	88.9	81.3	100	89.4
TJM(DeCAF) (Long et al. 2014)	89.3	89.1	79.6	77.8	84.7	78.3	90.8	79.6	97.9	88.4	78.6	100	86.2

methods across various adaptation scenarios, providing insights into the strengths and weaknesses of each algorithm in various domains.

A radar chart in Fig. 24 depicts the outcomes on the Office-Caltech dataset to enhance the readability of the results. This visual representation provides a comprehensive overview of the algorithmic performance, allowing for a more accessible and insightful comparison of results.

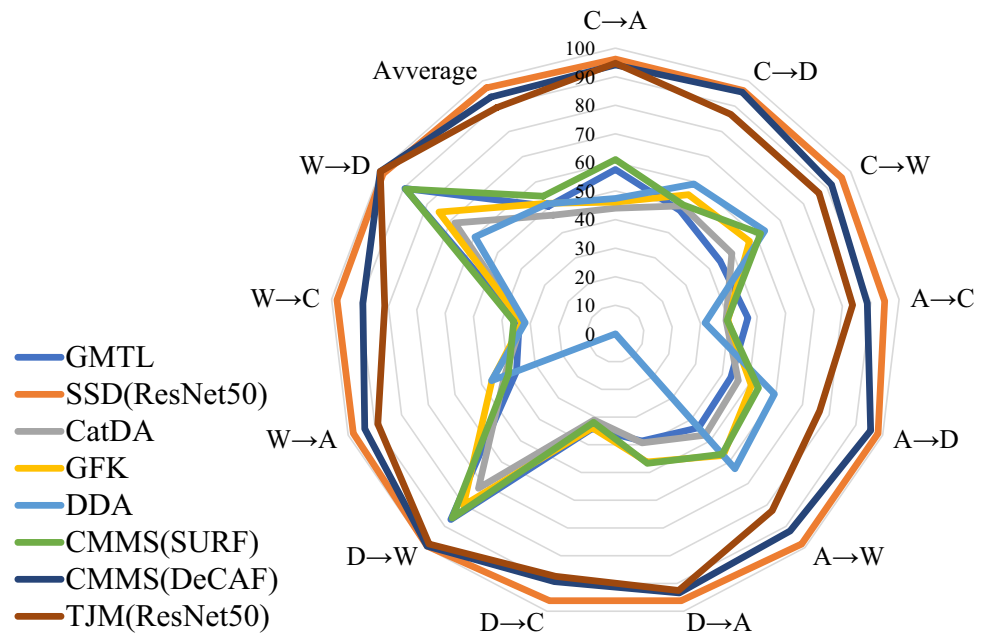
18 Conclusion

This review has examined the critical role of knowledge transfer in enhancing machine learning performance across diverse applications, particularly in scenarios where training and testing data do not share a common domain or distribution. Exploring various transfer learning methodologies, including single-source/single-target, single-source/multi-target, and multi-source/single-target frameworks, highlights the complexities and challenges inherent in real-world data scenarios. Robust techniques effectively leveraging knowledge from multiple sources are essential, especially in incomplete data cases. Furthermore, the evolving landscape of transfer learning, particularly with integrating fuzzy systems to manage uncertainty, demonstrates significant potential for improving system performance in varied contexts. Our systematic categorization of current approaches provides a foundational framework for researchers, facilitating a deeper understanding of the interrelationships among different methodologies and their implications for practical applications.

While the advancements in transfer learning are promising, several avenues for future research can further enhance its efficacy. One key direction involves the integration of unsupervised learning techniques, which could enable the harnessing of unlabeled data more effectively. This hybrid approach has the potential to mitigate challenges posed by limited labeled samples, allowing models to learn from the vast amounts of available unlabeled data. Additionally, developing adaptive transfer learning frameworks that can dynamically adjust to changing distributions over time would significantly improve model robustness, particularly in environments where data characteristics evolve, ensuring sustained performance.

Another promising research area is the development of real-time transfer learning systems. Such systems could facilitate immediate knowledge transfer from source to target domains, especially in applications like autonomous systems or online learning, allowing for continuous adaptation and learning as new data arrives. Moreover, exploring methods that enable models to generalize knowledge across a broader spectrum of domains, rather than relying on specific source-target pairs, would enhance the applicability of

Fig. 24 Comparison Results on Office-Caltech Dataset



transfer learning. This could involve creating more generalized feature extraction techniques and identifying domain-invariant representations.

Funding No funding was received for this study.

Data availability Data are included in the manuscript.

Declarations

Conflict of interests The authors declare that they do not have any conflict of interest.

Ethical approval This article does not contain any studies with human participants or animals performed by any authors.

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